CONFIDENTIAL

CLINICAL STUDY PROTOCOL AMENDMENT No. 2

STUDY TITLE:	A Phase 3, double-blind, multicenter, placebo-controlled study of PledOx used on top of modified FOLFOX6 (5-FU/FA and Oxaliplatin) to prevent chemotherapy induced peripheral neuropathy (CIPN) in the adjuvant treatment of patients with Stage III or high-risk Stage II colorectal cancer
SHORT TITLE:	**P**reventive treatment of **O**xa**L**iplatin induced peripheral neu**R**opathy in **A**djuvant colorectal cancer (**POLAR-A**)
PROTOCOL NUMBER:	PP06489
EUDRACT NUMBER:	2017-004707-43
IND NUMBER:	Not applicable
STUDY DRUG:	PledOx
SPONSOR:	PledPharma AB Grev Turegatan 11c SE 114 46 Stockholm Sweden Solasia Pharma K.K. 4F Sumitomo Fudosan Shiba-Koen Tower, 2-11-1, Shiba-koen, Minato-ku, Tokyo 105-0011, Japan (Co-sponsor in Asia)
CO-ORDINATING INVESTIGATORS:	Camilla Qvortrup, MD, PhD The Finsen Center – Rigshospitalet, Department of Oncology Blegdamsvej 9 Copenhagen, DK-2100 Denmark Kei Muro MD Director and Chief Department of Clinical Oncology Outpatient Treatment Center, Aichi Cancer Center Hospital, 1-1 Kanokoden, Chikusa-ku, Nagoya, Aichi 464-8681, Japan

PROTOCOL VERSION TABLE

Final, Version 4.0 incorporating Protocol Amendment 1	20 Jun 2018
Final, Version 5.0 incorporating Protocol Amendment 2	27 Sep 2018

STATEMENT OF CONFIDENTIALITY

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SIGNATURE PAGE

This study protocol was subjected to critical review. The information it contains is consistent with the current risk/benefit evaluation of the investigational product as well as with the moral, ethical and scientific principles governing clinical research as set out in the Declaration of Helsinki and the International Council for Harmonisation (ICH) guidelines on Good Clinical Practice (GCP).

This protocol is approved by:

CEO & President Nicklas Westerholm PledPharma AB	Signature	Sept 27, 2018 Date
Chief Medical Officer Stefan Carlsson PledPharma AB	Signature	Sept 27, 2018 Date
CEO & President Yoshihiro Arai Solasia Pharma K.K. (Co-sponsor in Asia)	Signature	Sep 28, 2018 Date
Principal Biostatistician Anders Persson Binomial Sweden AB	Signature	Sept 28, 2018 Date
Coordinating Investigator Camilla Qvortrup, MD, PhD The Finsen Center – Rigshospitalet, Denmark	Signature	Date
Coordinating Investigator Kei Muro, MD Aichi Cancer Center Hospital, Japan	Signature	Date

This Protocol amendment has been written in accordance with current ICH-GCP guidelines.

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Coordinating Investigator Camilla Qvortrup, MD, PhD The Finsen Center – Rigshospitalet, Denmark	Signature	26/9-18 Date
Coordinating Investigator Kei Muro, MD Aichi Cancer Center Hospital, Japan	Signature	Date

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Coordinating Investigator Camilla Qvortrup, MD, PhD The Finsen Center – Rigshospitalet, Denmark	Signature	Date
Coordinating Investigator Kei Muro, MD Aichi Cancer Center Hospital, Japan	Kei Muro Signature	28-Sep-2018 Date

This Protocol amendment has been written in accordance with current ICH-GCP guidelines.

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1. RATIONALE FOR PROTOCOL AMENDMENT

This substantial Protocol Amendment, dated 27 Sep 2018, includes the following changes to the final Protocol (Version 4.0, dated 20 Feb 2018):

- To include input from the Food and Drug Agency (FDA) to the protocol of study PP06498 (Polar M).
- To update background information.
- To correct the text and Schedule of Assessments for consistency and correctness.
- To amend for consistency and administrative changes (i.e. abbreviations, wording consistency, etc.; note: those changes are only given in this amendment in case they appear in the text parts shown for substantial amendments.)

2. SUMMARY OF CHANGES TO THE PROTOCOL

The following amendment applies to:

Section no. **Title**

Synopsis **Study Design**

Protocol text:

[…]

If a patient discontinues mFOLFOX6, the IMP should also be discontinued.

[…]

This will be replaced by:

[…]

If a patient discontinues ~~mFOLFOX6, the~~ **oxaliplatin, treatment with 5-FU/folinate and** IMP ~~should also~~ **may** be ~~discontinued~~ **continued**.

[…]

Reason:

To clarify procedures in case discontinuation of oxaliplatin treatment.

The following amendment applies to:

Section no. **Title**

Synopsis **Exploratory Objectives**

Protocol text:

To compare each dose of PledOx vs placebo regarding the following:

- Chronic CIPN by supporting analysis using the full FACT/GOG NTX-13.
- The cumulative dose of 5-FU during chemotherapy
- For both oxaliplatin and 5-FU: Dose intensity, number of cycles, dose reductions, reason(s) for dose reductions, patients with dose delays, and length of dose delays
- Quality of Life/health status
- The functional impairment (in the non-dominant hand) during long-term follow-up
- The worst pain in hands or feet during long-term follow-up
- Health economic impact

This will be replaced by:

To compare each dose of PledOx vs placebo regarding the following:

- Chronic CIPN by supporting analysis using the full FACT/GOG NTX-13.
- The cumulative dose of 5-FU during chemotherapy
- For both oxaliplatin and 5-FU: Dose intensity, number of cycles, dose reductions, reason(s) for dose reductions, patients with dose delays, and length of dose delays
- ~~Quality of Life/health status~~
- The functional impairment (in the non-dominant hand) during long-term follow-up
- The worst pain in hands or feet during long-term follow-up
- **Quality of life (QoL)/health status**
- Health economic impact

Reason:

To adjust exploratory objectives sequence for priority.

The following amendment applies to:

Section no. **Title**

Synopsis **Principal Selection Criteria**

Protocol text:

Inclusion criteria

[…]

4. The patient has undergone curative (R0) surgical resection performed not more than 8-12 weeks prior to randomization.

5. The patient has a postsurgical carcinoembryonic antigen (CEA) level ≤1.5 x upper limit of normal (ULN, in current smokers, CEA level ≤2.0 x ULN is allowed).

6. No prior anti-cancer therapy for CRC except radiotherapy or concomitant chemo-radiotherapy using a fluoropyrimidine alone for locoregional rectal cancer.

7. Patient indicated for up to 6 months of oxaliplatin-based chemotherapy and without any clinically observed neurological disorders.

[…]

Exclusion Criteria

[…]

4. Any prior oxaliplatin-based chemotherapy <1 year before the randomization.

[…]

This will be replaced by:

Inclusion criteria

[…]

4. The patient has undergone curative (R0) surgical resection performed ~~not more than 8-~~ **within** 12 weeks prior to randomization.

5. The patient has a postsurgical carcinoembryonic antigen (CEA) level ≤1.5 x upper limit of normal (ULN, in current smokers, CEA level ≤2.0 x ULN is allowed).

6. No prior anti-cancer therapy for CRC except radiotherapy or concomitant chemo-radiotherapy using a fluoropyrimidine alone for locoregional rectal cancer.

7. Patient indicated for up to 6 months of oxaliplatin-based chemotherapy and without ~~any clinically observed neurological disorders~~ **pathological findings of a neurologic exam performed prior to oxaliplatin treatment according to local practice**.

[…]

Exclusion Criteria

[…]

~~4. Any prior oxaliplatin-based chemotherapy <1 year before the randomization.~~

[…]

Reason:

To clarify inclusion and exclusion criteria following FDA input to study PP06498 (Polar M) and to correct time frame for tumor surgery.

The following amendment applies to:

Section no.	**Title**
Synopsis	**Criteria for Evaluation**

Protocol text:

[…]

Safety

- DFS at 12 and 24 months
- Safety and tolerability as assessed by adverse events (AEs), laboratory variables and vital signs

[…]

Exploratory Endpoints

- Frequency and proportion of patients with any CIPN scoring 1, 2, 3 or 4, in any of the first 4 items of the FACT/GOG-NTX-13 (i.e. FACT/GOG-NTX-4), targeting numbness, tingling or discomfort in hands and/or feet, at all other timepoints after the first dose of IMP

[…]

This will be replaced by:

[…]

Safety

- DFS at 12 and 24 months
- Safety and tolerability as assessed by adverse events (AEs), laboratory variables and vital signs
- **Proportion of patients with PledOX toxicities in addition to chemotherapy-related toxicities graded using the NCI-CTCAE v4.03**

[…]

Exploratory Endpoints

- ~~Frequency and p~~ **P**roportion of patients with any CIPN scoring 1, 2, 3 or 4, in any of the first 4 items of the FACT/GOG-NTX-13 (i.e. FACT/GOG-NTX-4), targeting numbness, tingling or discomfort in hands and/or feet, at all other timepoints after the first dose of IMP.

[…]

Reason:

To update endpoints following FDA input to study PP06498 (Polar M).

The following amendment applies to:

Section no.	**Title**
Synopsis	**Schedule of Assessments**

For ease of comprehension of the study and its design, a short version of the Schedule of Assessments per Appendix 1 was added to the synopsis. This short version can be found overleaf.

Schedule of Assessments (short version)

POLAR-A	SELECTION PHASE				TREATMENT PHASE													EOT		FOLLOW-UP PHASE			
					+/- 2 days													D 14 of last cycle +/- 3 days		+/- 1 week			
Visits in **Month**												**M3**							**M6**[11]	**M9**	**M12**	**M18**	**M24**
Visit in **Days** per **Cycle**	**D-28**	**D-14**	**D-7**	**D0**	**D1C1**	**D1C2**	**D1C3**	**D1C4**	**D1C5**	**D1C6**	**D1C7**		**D1C8**	**D1C9**	**D1C10**	**D1C11**	**D1C12**	**D14C12**					**EOS**
Visits in **Weeks**	**W-4**		**W-1**	**not a visit**	**W0**	**W2**	**W4**	**W6**	**W8**	**W10**	**W12**		**W14**	**W16**	**W18**	**W20**	**W22**	**W24**		**W36**	**W48**	**W72**	**W96**
Informed Consent	x																						
Demography, prior medication (including D-30 medication)			x																				
Medical history			x																				
ECG (QTcF)			x																				
Inclusion/exclusion criteria, randomization[6]				x																			
CT/MRI scan and disease assessment[1]	x																				x		x
DFS												x							x	x	x	x	x
CEA			x															x			x		x
Vital signs (T°, HR, BP) and weight (except after EOT)			x		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Pregnancy test (serum)[2]			x	results	x		x		x		x			x		x		x					
Hematology[3]			x	results	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x				
Biochemistry[4]			x	results	x		x		x		x			x		x		x					
Mn blood sample[5]		x		results				x					x				x	x					
ECOG Performance Status			x		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Grooved PEG-board & vibration sensitivity testing					x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
FACT/GOG-NTX-13[7]					x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Pain Numeric Rating Scale (NRS)[7]					x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
QoL (EQ-5D-5L)[7]					x			x				x	x				x	x	x	x	x	x	x
Cold Sensitivity Diary (pre-dose, post dose of D1 + D2 and D3)			x			x		x					x										
PledOx/placebo infusion (before FOLFOX)[8]					x	x	x	x	x	x	x		x	x	x	x	x						
Physical examination			x									x						x	x	x	x	x	x
MRI of CNS and blood Mn incl. neurological exam [10]					(x)	(x)	(x)	(x)	(x)	(x)	(x)	(x)	(x)	(x)	(x)	(x)	(x)	(x)					
Concomitant medication					x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
AEs[9]					x	x	x	x	x	x	x		x	x	x	x	x	x					
Health economics[12]					x														x		x	x	x

AE = adverse event; BP = blood pressure; CEA = Carcinoembryonic antigen; CNS = central nervous system; CT = computed tomography; D = day; DFS = disease free survival; ECG = electrocardiogram; ECOG = Eastern Cooperative Oncology Group; EoS = end of study; EoT = end of treatment; FACT/GOG-NTX-13 = Functional Assessment of Cancer Therapy/Gynecologic Oncology Group-Neurotoxicity-13-item subscale; HR = heart rate; M = month; Mn = manganese; MRI = magnetic resonance imaging; NRS = numeric rating scale; QoL = quality of life; T = temperature; W = week

1 CT-scan or MRI, but the same assessment should be performed throughout the study for each patient. For screening, results from a scan performed within 42 days prior to surgery or after surgery are allowed. Tumors will be analyzed and records will be evaluated locally.

2 WOCBP must give a negative serum pregnancy test at screening and then every 4 weeks while patient is on treatment

3 Hematology (every Treatment Visit): WBC count differential, ANC, RBC count, platelet count, and hemoglobin. Analyzed by a local hospital laboratory.

[4] Biochemistry (every 2nd Treatment Visit): ALP, total bilirubin, ALT, AST, serum creatinine, and albumin. Analyzed by a local hospital laboratory.

[5] Blood Mn (full blood): Collected at Treatment Visit 4, 8, 12 and EOT. Mn samples will also be taken whenever Parkinson-like symptoms occur. Analyzed by a central laboratory.

[6] Randomization – this is not a patient visit. This is the moment when the Investigator will confirm that a patient complies with all inclusion/exclusion criteria and the patient will be randomized into a treatment arm using the electronic randomization system within the eCRF. The Investigator will then request the unblinded pharmacist/designee to prepare the study treatment according to the randomization schedule available only to the unblinded hospital staff.

[7] Questionnaire to be completed electronically by a patient on a tablet in the hospital.

[8] IMP (PledOx or placebo) will be administered before the background chemotherapy (mFOLFOX6). Study treatment will be stopped in case of mFOLFOX6 treatment discontinuation, disease progression, intolerable toxicity or withdrawal of consent.

[9] All AEs including SAEs will be collected up to 30 days after the last administration of IMP and followed until resolution.

[10] In patients who develop Parkinson-like symptoms, a neurological examination and an MRI of the CNS will be performed and an additional blood Mn sample will be analyzed. An MRI of the CNS should also be performed if elevated Mn is found during the scheduled Mn samples.

[11] The Assessment Visit at Month 6 will be combined with the EOT Visit only if a patient had the last study treatment dose on Treatment Visit 12 (Week 22).

[12] The Health Economic Impact Questionnaire will be completed by the Investigator.

The following amendment applies to:

Section no.	**Title**
Section 3.4	**Non-Clinical Findings**

Protocol text:

The non-clinical safety program conducted with calmangafodipir consisted of pilot 4-week and clinical trial-enabling 13-week repeat dose toxicity studies in rats and dogs. The latter included assessment of safety pharmacology endpoints. In order to support the use of the non-clinical safety information on mangafodipir, this compound was included as a reference compound in both the rat and dog 13-weeks toxicity studies.

In the 13-week studies, animals were dosed 3 times a week; 39 administrations in total. Rats given the high calmangafodipir dose were administered 12-, 72- and 39-fold the individual, the fortnightly and the cumulative clinical dose (based on a 12-cycle regimen), respectively, on a mg/m2 basis. High dose dogs were administered 34-, 208- and 113-fold the individual, the fortnightly, and the cumulative clinical dose on a mg/m2 basis. No indication of any central nervous system or respiratory system effects was noted in rats after dosing with up to 12-fold the recommended clinical dose of 5 µmol/kg and there were no indications of adverse effects on cardiovascular function (blood pressure and electrocardiogram [ECG]) in dogs treated with i.v. doses of up to 34-fold the recommended clinical dose on a mg/m2 basis. High dose rats showed slightly lower absolute and/or relative weights of testes and epididymides and slight weight reductions in the pituitary of all male groups and the high female dose group. The mechanism(s) underlying these observations is not clear. However, there were no clear pathological findings in these organs and their organ weights were similar to controls at the end of the recovery period. There were no calmangafodipir-related microscopic findings. Dogs administered calmangafodipir showed a dose and time-related loss of fur as well as hypersalivation and/or vomiting during the infusion with the high calmangafodipir dose. Body weight and body weight gain mean values were lower than controls in high dose males and females but showed evidence of recovery during the treatment free period. Histological examination showed testicular tubular atrophy in one high dose animal. Chronic toxicity studies in rats (6 months) and dogs (9 months) have been completed.

The aim of the 26-week intravenous toxicity study in Wistar rats with 28-day recovery period study was to obtain information on the toxicity of the test item in Wistar rats when administered at four dose levels of 0, 40, 120, 360 and 1440 µmol/kg body weight (8, 24, 72 and 288 fold of the assumed clinical dose on a mg/kg basis and 1.2, 3.8, 11.6 and 46 fold of the clinical dose on mg/m2 basis) once every second week via short intravenous infusion at a rate of 0.6 mL/minute, for 26 weeks. The study was performed to support formal clinical trial. Reversibility of any changes was assessed after a 28-day recovery period. Blood samples were collected for assessment of exposure to Mn and fodipir and its dephosphorylated metabolites. One control TK animal died early during the study, a clear cause of death could not be established. There were no adverse clinical signs recorded during the study or any adverse effects in Neurological Assessment (FOB) in any of the groups. There were no test item related effects on body weight, body weight gain, food consumption, vaginal smears and ophthalmology and urine parameters. There were test item related but non-adverse findings in hematology, clinical chemistry parameters. There were no test item related effects observed at necropsy, in organ weight data and in histopathology. In conclusion, under the condition of this study, the treatment with calmangafodipir once every second week via short intravenous

infusion at a rate of 0.6 mL/minute, for 26 weeks to Wistar rats, did not cause adverse effects. Therefore, the NOAEL (no-observed-adverse-effect-level) in this study was determined to be 1440 µmol/kg body weight (highest dose) for both sexes.

The aim of the 9 month chronic toxicity study in dogs was to obtain information on the toxicity of the test item in Beagle dogs when administered at 0, 30, 100, 300 and 1400 µmol/kg body weight (equivalent to 0, 22.2, 74.1, 222.1 and 1037 mg/kg bw) once every second week via short intravenous infusion at a rate of 1.2 mL/minute, for 39 weeks. The study was performed to support formal clinical trial. Reversibility of changes were assessed after a 28-day recovery period. Blood samples were collected for bioanalysis to provide information on the systemic exposure to fodipir and its dephosphorylated metabolites and to whole blood manganese concentration.

There was 1 mortality during the study during the treatment period. This high dose male was euthanized on Day 127 (on the day of the 10th infusion) due to moderate vomiting and presence of vocalization, clonic convulsion, aggressiveness, increased salivation, involuntary defecation and urination.

NOAEL in the 9-months toxicity study in the dog was the mid high dose tested, i.e. 300 µmol/kg and that the lowest-observed-adverse-effect level (LOAEL) was the highest dose investigated (1400 µmol/kg). The findings considered adverse in high dose animals consisted of slight to moderate vomiting during the days of infusion and increased salivation, also seen on treatment days, after about 3 months of treatment. Male animals of the high dose had 25 55% lower food consumption compared to controls and consequently had reduced body weight and body weight gain. ECG-monitoring, taken approximately 0.5 hours post treatment, showed a consistently lower heart rate in high dose male dogs. This effect was no longer seen after a 28 days recovery period. Since the NOAEL in this 9 months study was 33-fold higher than the highest clinical dose, on a dose/body surface area basis, it is concluded that the dog data does not indicate any risk for treatment related toxicities. Importantly, the effects seen in dogs at the LOAEL (155-times higher than the highest clinical dose) were not serious as referenced above.

Fertility studies in male rats and embryofetal development studies in rats and rabbits have been conducted with mangafodipir. Skeletal malformations were observed in the embryofetal development study in rats at the lowest dose investigated and embryofetal toxicity was observed in rabbits. Identical effects were seen after administration of equimolar MnCl2. It is the Sponsor's opinion that mangafodipir and calmangafodipir should have a similar reproduction toxicity liability.

Calmangafodipir was investigated for in vitro genotoxicity using the Ames Salmonella Assay and a micronucleus assay in human lymphocytes. No mutagenic activity was recorded. In the in vitro micronucleus assay with an extended 24 + 24 hour treatment at a cytotoxic dose (reduction in replication index of 66%), a weak induction of micronuclei was observed. The effect observed at such a high cytotoxic concentration is of questionable biological relevance. When the clastogenic activity of calmangafodipir was examined in a rat bone marrow micronucleus study in rats, no such activity was observed when after the administration of up to 2000 mg/kg (the regulatory maximum dose) on two consecutive days and bone marrow sampling 24 hours after the last dose.

This will be replaced by:

~~The non-clinical safety program conducted with calmangafodipir consisted of pilot 4-week and clinical trial-enabling 13-week repeat dose toxicity studies in rats and dogs. The latter included assessment of safety pharmacology endpoints. In order to support the use of the non-clinical~~

~~safety information on mangafodipir, this compound was included as a reference compound in both the rat and dog 13-weeks toxicity studies.~~

~~In the 13-week studies, animals were dosed 3 times a week; 39 administrations in total. Rats given the high calmangafodipir dose were administered 12-, 72- and 39-fold the individual, the fortnightly and the cumulative clinical dose (based on a 12- cycle regimen), respectively, on a mg/m2 basis. High dose dogs were administered 34-, 208- and 113-fold the individual, the fortnightly, and the cumulative clinical dose on a mg/m2 basis. No indication of any central nervous system or respiratory system effects was noted in rats after dosing with up to 12-fold the recommended clinical dose of 5 µmol/kg and there were no indications of adverse effects on cardiovascular function (blood pressure and electrocardiogram [ECG]) in dogs treated with i.v. doses of up to 34-fold the recommended clinical dose on a mg/m2 basis. High dose rats showed slightly lower absolute and/or relative weights of testes and epididymides and slight weight reductions in the pituitary of all male groups and the high female dose group. The mechanism(s) underlying these observations is not clear. However, there were no clear pathological findings in these organs and their organ weights were similar to controls at the end of the recovery period. There were no calmangafodipir-related microscopic findings. Dogs administered calmangafodipir showed a dose and time-related loss of fur as well as hypersalivation and/or vomiting during the infusion with the high calmangafodipir dose. Body weight and body weight gain mean values were lower than controls in high dose males and females but showed evidence of recovery during the treatment-free period. Histological examination showed testicular tubular atrophy in one high dose animal. Chronic toxicity studies in rats (6 months) and dogs (9 months) have been completed.~~

~~The aim of the 26-week intravenous toxicity study in Wistar rats with 28-day recovery period study was to obtain information on the toxicity of the test item in Wistar rats when administered at four dose levels of 0, 40, 120, 360 and 1440 µmol/kg body weight (8, 24, 72 and 288 fold of the assumed clinical dose on a mg/kg basis and 1.2, 3.8, 11.6 and 46 fold of the clinical dose on mg/m2 basis) once every second week via short intravenous infusion at a rate of 0.6 mL/minute, for 26 weeks. The study was performed to support formal clinical trial. Reversibility of any changes was assessed after a 28-day recovery period. Blood samples were collected for assessment of exposure to Mn and fodipir and its dephosphorylated metabolites. One control TK animal died early during the study, a clear cause of death could not be established. There were no adverse clinical signs recorded during the study or any adverse effects in Neurological Assessment (FOB) in any of the groups. There were no test item related effects on body weight, body weight gain, food consumption, vaginal smears and ophthalmology and urine parameters. There were test item related but non-adverse findings in hematology, clinical chemistry parameters. There were no test item related effects observed at necropsy, in organ weight data and in histopathology. In conclusion, under the condition of this study, the treatment with calmangafodipir once every second week via short intravenous infusion at a rate of 0.6 mL/minute, for 26 weeks to Wistar rats, did not cause adverse effects. Therefore, the NOAEL (no-observed-adverse-effect-level) in this study was determined to be 1440 µmol/kg body weight (highest dose) for both sexes.~~

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~~exposure to fodipir and its dephosphorylated metabolites and to whole blood manganese concentration.~~

~~There was 1 mortality during the study during the treatment period. This high dose male was euthanized on Day 127 (on the day of the 10th infusion) due to moderate vomiting and presence of vocalization, clonic convulsion, aggressiveness, increased salivation, involuntary defecation and urination.~~

~~NOAEL in the 9-months toxicity study in the dog was the mid high dose tested, i.e. 300 µmol/kg and that the lowest-observed-adverse-effect level (LOAEL) was the highest dose investigated (1400 µmol/kg). The findings considered adverse in high dose animals consisted of slight to moderate vomiting during the days of infusion and increased salivation, also seen on treatment days, after about 3 months of treatment. Male animals of the high dose had 25 55% lower food consumption compared to controls and consequently had reduced body weight and body weight gain. ECG-monitoring, taken approximately 0.5 hours post treatment, showed a consistently lower heart rate in high dose male dogs. This effect was no longer seen after a 28 days recovery period. Since the NOAEL in this 9 months study was 33-fold higher than the highest clinical dose, on a dose/body surface area basis, it is concluded that the dog data does not indicate any risk for treatment related toxicities. Importantly, the effects seen in dogs at the LOAEL (155-times higher than the highest clinical dose) were not serious as referenced above.~~

Results from pharmacology studies show that calmangafodipir protects BALB/c mice against the myelosuppressive effects of oxaliplatin and that the analogue mangafodipir effectively prevents the onset of locomotor and sensory disturbances as well as neuromuscular hyperexcitability in C57Bl/6 mice treated with oxaliplatin. Importantly, calmangafodipir does not interfere negatively with the antitumor activity of oxaliplatin; on the contrary, preclinical data suggest that it increases the antitumor efficacy in immune competent mice which might be an inherent property of DPDP or its dephosphorylated counterparts.

The general toxicity program with calmangafodipir consists of pilot repeat dose studies of 28 days duration in rats and dogs and clinical trial enabling 13-week studies in the same species as well as a 6 months study in rats and a 9 months study in dogs. The 28-day and 13-weeks studies used a dosage regimen of three injections per week rather than the once fortnightly regimen used in humans. The 6 and 9 months studies used the clinical regimen of once every fortnight.

Overall, the rat 13-week study revealed no toxicities of significant clinical concern in spite of single dose levels of calmangafodipir being up to 12-fold higher than the highest proposed clinical dose (5 µmol/kg) and 72-fold higher based on a fortnightly dose both calculated on a mg/m2 basis. Calmangafodipir-related findings were restricted to apparent weight reductions of the testes and epididymides in high dose animals and lower pituitary weights in all male treated animals and in high dose females. The good tolerability observed in the 13-week study was confirmed in the 6 months study where the no-observed-adverse-effect-level (NOAEL) was the highest dose investigated; 1440 µmol/kg (1066.6 mg/kg body weight). This NOAEL is 46-fold the clinical dose on a mg/m2 basis. Exposure measurement in terms of whole blood Mn levels as well as C_{max} and $AUC_{0\text{-}inf}$ PLED give additional reassurance of good tolerability relative to the corresponding clinical exposures.

The dog 13-week study also did not reveal any toxicities of significant clinical concern in spite of single dose levels of calmangafodipir being 34-fold higher than the highest clinical

dose (5 µmol/kg) on a mg/m2 basis and 204-fold the cumulative dose over the clinical dosing interval of 14 days. The treatment could possibly have affected estrus cycling but owing to the low age of the animals the relationship of this to the treatment is uncertain. One high dose calmangafodipir animal showed testicular tubular atrophy. In addition to this, all animals showed various injections site reactions with a possibly higher incidence of hemorrhage in treated animals than in the controls, a finding of questionable clinical relevance in view of the absence of any calmangafodipir related injection site reactions in the 9 months dog study. This study showed vomiting and salivation in high dose animals of both sexes, lowered food consumption in high dose males and one early sacrifice of a high dose male owing to poor clinical condition. The NOAEL was the mid high dose, a dose being 33-fold higher than the clinical dose on a mg/m2 basis. Results from exposure measurement (whole blood Mn levels as well as C_{max} and $AUC_{0\text{-}inf}$ PLED) relative to that in man, provide reassurance of good tolerability of calmangafodipir in the dog.

Fertility studies in male rats and embryofetal development studies in rats and rabbits have been conducted with mangafodipir. Skeletal malformations were observed in the embryofetal development study in rats at the lowest dose investigated and embryofetal toxicity was observed in rabbits. Identical effects were seen after administration of equimolar MnCl2. It is the Sponsor's opinion that mangafodipir and calmangafodipir should have a similar reproduction toxicity liability.

Calmangafodipir was investigated for in vitro genotoxicity using the Ames Salmonella Assay and a micronucleus assay in human lymphocytes. No mutagenic activity was recorded. In the in vitro micronucleus assay with an extended 24 + 24 hour treatment at a cytotoxic dose (reduction in replication index of 66%), a weak induction of micronuclei was observed. The effect observed at such a high cytotoxic concentration is of questionable biological relevance. When the clastogenic activity of calmangafodipir was examined in a rat bone marrow micronucleus study in rats, no such activity was observed when after the administration of up to 2000 mg/kg (the regulatory maximum dose) on two consecutive days and bone marrow sampling 24 hours after the last dose.

Reason:

To updated non-clinical background information with most current data.

The following amendment applies to:

Section no.	**Title**
Section 3.5	**Clinical Findings**

Protocol text:

Oxaliplatin causes disabling acute and chronic peripheral neuropathy. We explored the preventive effects of calmangafodipir, mimicking the mitochondrial enzyme manganese superoxide dismutase, thereby protecting cells from oxidative stress, in a placebo-controlled, double-blinded randomized Phase 2 study (ClinicalTrials.gov, NCT01619423) in patients with metastatic colorectal cancer (mCRC).

mCRC patients treated with mFOLFOX6 (folinic acid 200 mg/m^2, 5-FU bolus 400 mg/m^2, oxaliplatin 85 mg/m^2 and 5-FU 2400 mg/m^2 continuous infusion for 46 hours) every fortnight for 8 cycles in first or second line were eligible. Calmangafodipir was given in a Phase 1 dose-finding (Part 1) and in a Phase 2 placebo-controlled part of the study (Part 2 and 3), as a 5 minutes infusion, 10 minutes prior to oxaliplatin. Neurotoxicity was evaluated by the physician

using the Oxaliplatin Sanofi Specific Scale (OSSS) and by the patient using the cold allodynia test (Ventzel cylinder [Ventzel et al. 2016]) and the Leonard scale (oxaliplatin-associated neuropathy questionnaire, OANQ [Leonard et al. 2005]).

Eleven patients were included in Phase 1 without any detectable toxicity to calmangafodipir. In the Phase 2 study, 173 patients were randomized to placebo (n=60), calmangafodipir 2 µmol/kg (n=57) and calmangafodipir 5 µmol/kg (n=45, (part 2b) initially 10 µmol/kg, n=11 (part 2a). Calmangafodipir-treated patients (all three doses pooled) had less physician graded neurotoxicity (odds ratio (90% confidence interval one-sided upper level) 0.615 (1.159), p=0.158), less problems with cold allodynia and fewer sensory symptoms in the Leonard scale during treatment and during follow-up 3 and 6 months after last dose. Response rate, progression-free and overall survival did not differ among groups.

Calmangafodipir at a dose of 5 µmol/kg appears to prevent the development of oxaliplatin-induced acute and delayed CIPN without apparent influence on tumor outcomes.

[…]

<u>This will be replaced by:</u>

Oxaliplatin causes disabling acute and chronic peripheral neuropathy. We explored the preventive effects of calmangafodipir, mimicking the mitochondrial enzyme manganese superoxide dismutase, thereby protecting cells from oxidative stress, in a placebo-controlled, double-blinded randomized Phase 2 study (ClinicalTrials.gov, NCT01619423) in patients with metastatic colorectal cancer (mCRC).

mCRC patients treated with mFOLFOX6 (folinic acid 200 mg/m^2, 5-FU bolus 400 mg/m^2, oxaliplatin 85 mg/m^2 and 5-FU 2400 mg/m^2 continuous infusion for 46 hours) every fortnight for 8 cycles in first or second line were eligible. Calmangafodipir was given in a Phase 1 dose-finding (Part 1) and in a Phase 2 placebo-controlled part of the study (Part 2 and 3), as a 5 minutes infusion, 10 minutes prior to oxaliplatin. Neurotoxicity was evaluated by the physician using the Oxaliplatin Sanofi Specific Scale (OSSS) and by the patient using the cold allodynia test (Ventzel cylinder [Ventzel et al. 2016]) and the Leonard scale (oxaliplatin-associated neuropathy questionnaire, OANQ [Leonard et al. 2005]).

Eleven patients were included in Phase 1 without any detectable toxicity to calmangafodipir. In the Phase 2 study, 173 patients were randomized to placebo (n=60), calmangafodipir 2 µmol/kg (n=57) and calmangafodipir 5 µmol/kg (n=45, (part 2b) initially 10 µmol/kg, n=11 (part 2a). Calmangafodipir-treated patients (all three doses pooled) had less physician graded neurotoxicity (odds ratio (90% confidence interval one-sided upper level) 0.615 (1.159), p=0.158), less problems with cold allodynia and fewer sensory symptoms in the Leonard scale during treatment and during follow-up 3 and 6 months after last dose. Response rate, progression-free and overall survival did not differ among groups.

Calmangafodipir at a dose of 5 µmol/kg appears to prevent the development of oxaliplatin-induced acute and delayed CIPN without apparent influence on tumor outcomes.

SUNCIST a randomized, double-blind, placebo-controlled, ascending (2, 5 and 10 µmol/kg vs placebo, sequential enrolling cohort, single-dose study in healthy Japanese and Caucasian men, age 20 to 55 years, inclusive. 48 participants (24 Japanese and 24 Caucasian) were studied in two groups of three sequentially enrolling dose cohorts. Group 1 consisted of three cohorts of 8 Japanese subjects each, and Group 2 consisted of 3 cohorts of 8 Caucasian subjects each. Calmangafodipir at doses up to 10 µmoL/kg was safe and well tolerated. Calmangafodipir exposure ($AUC_{0\text{-last}}$ and C_{max}) was comparable

for the Japanese and Caucasian subjects. $AUC_{0\text{-}last}$ and C_{max} increased proportionately with increasing dose.

[…]

Reason:

To update clinical background data.

The following amendment applies to:

Section no.	**Title**
Section 3.9	**Population**

Protocol text:

The target population for this study is patients with pathologically confirmed CRC including Stage III carcinoma (any T N1,2 M0) or Stage II carcinoma (T3,4 N0 M0). Patient must have undergone curative surgical treatment, generally not more than 8-12 weeks prior to randomization.

Approximately 280 patients will be enrolled. The planned population of 280 patients will be enrolled at approximately 95 study centers.

There will be no additional patients included in order to compensate for withdrawals.

This will be replaced by:

The target population for this study is patients with pathologically confirmed CRC including Stage III carcinoma (any T N1,2 M0) or Stage II carcinoma (T3,4 N0 M0). Patient must have undergone curative surgical treatment, ~~generally not more than 8-~~ **within** 12 weeks prior to randomization.

Approximately 280 patients will be enrolled. The planned population of 280 patients will be enrolled at approximately 95 study centers.

There will be no additional patients included in order to compensate for withdrawals.

Reason:

To correct time frame for tumor surgery.

The following amendment applies to:

Section no.	**Title**
Section 4.1.3	**Exploratory Objectives**

Protocol text:

To compare each dose of PledOx vs placebo regarding the following:

- Chronic CIPN by supporting analysis using the full FACT/GOG NTX-13.
- The cumulative dose of 5-FU during chemotherapy
- For both oxaliplatin and 5-FU: Dose intensity, number of cycles, dose reductions, reason(s) for dose reductions, patients with dose delays, and length of dose delays
- Quality of Life/health status
- The functional impairment (in the non-dominant hand) during long-term follow-up
- The worst pain in hands or feet during long-term follow-up
- Health economic impact

This will be replaced by:

To compare each dose of PledOx vs placebo regarding the following:

- Chronic CIPN by supporting analysis using the full FACT/GOG NTX-13.
- The cumulative dose of 5-FU during chemotherapy
- For both oxaliplatin and 5-FU: Dose intensity, number of cycles, dose reductions, reason(s) for dose reductions, patients with dose delays, and length of dose delays
- ~~Quality of Life/health status~~
- The functional impairment (in the non-dominant hand) during long-term follow-up
- The worst pain in hands or feet during long-term follow-up
- **QoL/health status**
- Health economic impact

Reason:

To update exploratory objectives for priority.

The following amendment applies to:

Section no.	**Title**
Section 5.1.2	**Secondary Endpoints**

Protocol text:

[…]

Safety

- DFS at 12 and 24 months
- Safety and tolerability as assessed by adverse events (AEs), laboratory variables and vital signs

This will be replaced by:

Safety

- DFS at 12 and 24 months
- Safety and tolerability as assessed by adverse events (AEs), laboratory variables and vital signs
- **Proportion of patients with PledOX toxicities in addition to chemotherapy-related toxicities graded using the National Cancer Institute Common Terminology Criteria for Adverse Events (NCI-CTCAE) v4.03**

Reason:

To update endpoints following FDA input to study PP06498 (Polar M).

The following amendment applies to:

Section no.	**Title**
Section 5.1.3	**Exploratory Endpoints**

Protocol text:

- Frequency and proportion of patients with any CIPN scoring 1, 2, 3 or 4, in any of the first 4 items of the FACT/GOG-NTX-13 (i.e. FACT/GOG-NTX-4), targeting numbness, tingling or discomfort in hands and/or feet, at all other timepoints after the first dose of IMP.

[…]

This will be replaced by:

- Frequency ~~and proportion~~ **Proportion** of patients with any CIPN scoring 1, 2, 3 or 4, in any of the first 4 items of the FACT/GOG-NTX-13 (i.e. FACT/GOG-NTX-4), targeting numbness, tingling or discomfort in hands and/or feet, at all other timepoints after the first dose of IMP.

[…]

Reason:

To update exploratory endpoints for consistency with safety endpoints.

The following amendment applies to:

Section no.	**Title**
Section 5.2	**Overall Study Design and Plan**

Protocol text:

[…]

t=0 (baseline: Day 1, Cycle 1); R=randomization; EoT=end of treatment; DFS=disease free survival

Treatment Phase

Patients will be treated with IMP (i.e. PledOx or placebo) on top of the background chemotherapy (mFOLFOX6), unless preceded by disease progression, intolerable toxicity or withdrawal of consent.

If a patient discontinues mFOLFOX6, the IMP should also be discontinued.

[…]

This will be replaced by:

[…]

t=0 (baseline: Day 1, Cycle 1); R=randomization; EoT=end of treatment; DFS=disease free survival

Note: CIPN will be evaluated by the first 4 items of the FACT/GOG-NTX-13 (i.e. FACT/GOG-NTX-4), targeting numbness, tingling or discomfort in hands and/or feet, also at 9, 12, 18 and 24 months after the first dose of IMP

Treatment Phase

Patients will be treated with IMP (i.e. PledOx or placebo) on top of the background chemotherapy (mFOLFOX6), unless preceded by disease progression, intolerable toxicity or withdrawal of consent.

If a patient discontinues ~~mFOLFOX6, the~~ **oxaliplatin, treatment with 5-FU/folinate and** IMP ~~should also~~ **may be** ~~discontinued~~ **continued**.

[…]

Reason:

To update chemotherapy treatment adjustments following FDA input to study PP06498 (Polar M).

The following amendment applies to:

Section no.	**Title**
Section 5.2.1.1	**Screening Period**

Protocol text:

[…]

Within 7 days prior to randomization:

- Carcinoembryonic antigen (CEA)
- Medical history (including an assessment of conditions and symptoms present at baseline) and prior medications (including those medications taken 30 days prior to first dose of IMP)
- Physical examination (including height and weight measurements)
- Vital signs (including temperature, pulse rate, and blood pressure)
- Eastern Cooperative Oncology Group (ECOG) performance status
- Serum β-human choriongonadotropin pregnancy test in females of child-bearing potential
- Demography
- Hematology sample
- Serum biochemistry sample
- 12-lead ECG
- The cold sensitivity patient diary

This will be replaced by:

[…]

Within 7 days prior to randomization:

- Carcinoembryonic antigen (CEA)
- Medical history (including an assessment of conditions and symptoms present at baseline) and prior medications (including those medications taken 30 days prior to first dose of IMP)
- Physical examination (including height and weight measurements, **and a neurologic assessment performed prior oxaliplatin treatment according to local practice)**
- Vital signs (including temperature, pulse rate, and blood pressure)
- Eastern Cooperative Oncology Group (ECOG) performance status
- Serum β-human choriongonadotropin **(β-HCG)** pregnancy test in ~~females~~ **women** of ~~child-bearing~~ **childbearing** potential **(WOCBP)**
- Demography
- ~~Hematology sample~~
- ~~Serum biochemistry sample~~
- **Hematology sample to assess white blood cell (WBC) count differential, absolute neutrophil count (ANC), red blood cell (RBC) count, platelet count and hemoglobin (to be analyzed by a local laboratory)**
- **Serum biochemistry sample to assess ALP, total bilirubin, alanine aminotransferase (ALT), aspartate aminotransferase (AST), serum creatinine and albumin (to be analyzed by a local laboratory)**
- 12-lead ECG
- The ~~cold sensitivity patient~~ **Cold Sensitivity questionnaire (paper** diary**)**

Reason:

To update examinations following FDA input to study PP06498 (Polar M)and to harmonize wording regarding the Cold Sensitivity questionnaire.

The following amendment applies to:

Section no.	**Title**
Section 5.2.1.3	**Treatment Phase**

Protocol text:

Each treatment cycle of chemotherapy will last for 2 weeks.

Treatment Visit 1 (Week 0): Day 1 of Cycle 1 of mFOLFOX6 chemotherapy

Before IMP infusion

- Concomitant medications
- Vital signs and weight
- ECOG performance status
- Hematology sample collection (as per local practice)
- Biochemistry sample collection (as per local practice)
- Serum β-HCG pregnancy test in females of child-bearing potential
- Review of results of blood Mn sample
- FACT/GOG- NTX-13
- Grooved Pegboard
- Vibration sensitivity test, performed using a graduated tuning fork
- EQ-5D-5L questionnaire
- Pain NRS questionnaire
- Health economic impact
- AEs

IMP infusion

- PledOx or placebo will be administered by an i.v. infusion over approximately 5 minutes, 10 minutes prior to mFOLFOX6

After IMP infusion

- AEs

mFOLFOX6 infusion

Treatment Visits 2, 3, 4, 5, 6 (Weeks 2, 4, 6, 8, 10 ±2 days)

Assessments at these visits are similar to Treatment Visit 1 with the following exceptions:

- Biochemistry parameters are assessed every 4 weeks i.e. Treatment Visit 3 and 5 only
- Serum β-HCG pregnancy test in females of child-bearing potential will be performed every 4 weeks i.e. Treatment Visit 3 and 5 only
- Blood Mn, EQ-5D-5L, are assessed at Treatment Visit 4 only
- The cold sensitivity patient diary is completed before IMP infusion at Treatment Visits 2 and 4 and on days 1, 2, and 3 after IMP infusion
- Health economic impact data are not collected at these visits

Assessment Visit: Month 3 (±1 week)

Note: If the patient continues with the study treatment, Treatment Visit 7 will be combined with the Month 3 Assessment Visit. The Assessment Visit at Month 3 will be performed as a separate visit only if a patient stops study treatment prior to Treatment Visit 7 (Week 12).

The following assessments will be performed at each Assessment Visit:

- Concomitant medications
- Physical examination
- Vital signs
- ECOG performance status
- Hematology sample
- Biochemistry sample
- FACT/GOG-NTX-13
- Grooved pegboard
- Vibration sensitivity test, performed using a graduated tuning fork
- EQ-5D-5L
- Pain NRS questionnaire
- DFS

If a patient does not attend the hospital for the Assessment visit, the study staff will conduct a recorded telephone call with the patient to collect the relevant data. In such cases, study staff will read through the questionnaires (e.g. FACT/GOG-NTX-13, etc) and will enter the data into an electronic device/tablet according to the patient's response.

Treatment Visits 7, 8, 9, 10, 11, 12 (Weeks 12, 14, 16, 18, 20, 22 ±2 days)

Assessments at these visits are similar to Treatment Visit 1 with the following exceptions:

- Biochemistry parameters are assessed every 4 weeks i.e. Treatment Visit 7, 9 and 11 only
- Serum β-HCG pregnancy test in females of child-bearing potential will be performed every 4 weeks i.e. Treatment Visit 7, 9 and 11 only
- Blood Mn, and QoL/health status (EQ-5D-5L) are assessed at Treatment Visits 8 and 12 only
- The cold sensitivity patient diary is completed before IMP infusion at Treatment Visit 8 and on Days 1, 2, and 3 after the study treatment infusion is assessed only
- Health economic impact data are not collected at these visits

[…]

End of Treatment (EOT) Visit

The following assessments will be performed at the end of the last cycle of mFOLFOX6:

- CEA
- Concomitant medications
- Physical examination
- Vital signs
- ECOG performance status
- Hematology sample
- Biochemistry sample
- Blood Mn sample
- FACT/GOG-NTX-13
- Grooved pegboard
- Vibration sensitivity test, performed using a graduated tuning fork
- EQ-5D-5L

- Pain NRS questionnaire
- AEs

All AEs including SAEs will be collected up to 30 days after the last administration of IMP and followed until resolution.

This will be replaced by:

Each treatment cycle of chemotherapy will last for 2 weeks.

Treatment Visit 1 (Week 0): Day 1 of Cycle 1 of mFOLFOX6 chemotherapy

Before IMP infusion

- Concomitant medications
- Vital signs and weight
- ECOG performance status
- ~~Hematology sample collection (as per local practice)~~
- ~~Biochemistry sample collection (as per local practice)~~
- ~~Serum β-HCG pregnancy test in females of child-bearing potential~~
- **Check of hematology, biochemistry and serum β-HCG pregnancy test results from samples taken within 7 days prior to Treatment Visit 1 or prior to start of infusion at Treatment Visit 1**
- Review of results of blood Mn sample
- FACT/GOG- NTX-13
- Grooved Pegboard
- Vibration sensitivity test, performed using a graduated tuning fork
- EQ-5D-5L questionnaire
- Pain NRS questionnaire
- Health economic impact
- AEs

IMP infusion

- PledOx or placebo will be administered by an i.v. infusion over approximately 5 minutes, 10 minutes prior to mFOLFOX6

After IMP infusion

- AEs

mFOLFOX6 infusion

Treatment Visits 2, 3, 4, 5, 6 (Weeks 2, 4, 6, 8, 10 ±2 days)

Assessments at these visits are similar to Treatment Visit 1 with the following exceptions:

- Biochemistry parameters are assessed every 4 weeks i.e. Treatment Visit 3 and 5 only **(samples to be taken in advance of the visit to have the results available before the start of the new chemotherapy and IMP cycle)**
- Serum β-HCG pregnancy test in ~~females of child-bearing potential~~ **WOCBP** will be performed every 4 weeks i.e. Treatment Visit 3 and 5 only **(test to be taken in advance of the visit to have the results available before the start of the new chemotherapy and IMP cycle)**
- Blood Mn, EQ-5D-5L, PK sample and ECG (at pre-selected sites in ~~72~~ **84** patients only) are assessed at Treatment Visit 4 only
- The ~~cold sensitivity patient~~ **Cold Sensitivity questionnaire (paper** diary**)** is completed before IMP infusion at Treatment Visits 2 and 4 ~~and~~ on days 1, 2, and 3 after IMP infusion

- Health economic impact data are not collected at these visits

[…]

Assessment Visit: Month 3 (±1 week)

Note: If the patient continues with the study treatment, Treatment Visit 7 will be combined with the Month 3 Assessment Visit. The Assessment Visit at Month 3 will be performed as a separate visit only if a patient stops study treatment prior to Treatment Visit 7 (Week 12).

The following assessments will be performed at each Assessment Visit:

- Concomitant medications
- Physical examination
- Vital signs
- ECOG performance status
- Hematology sample
- ~~Biochemistry sample~~
- FACT/GOG-NTX-13
- Grooved pegboard
- Vibration sensitivity test, performed using a graduated tuning fork
- EQ-5D-5L
- Pain NRS questionnaire
- DFS

~~If a patient does not attend the hospital for the Assessment visit, the study staff will conduct a recorded telephone call with the patient to collect the relevant data. In such cases, study staff will read through the questionnaires (e.g. FACT/GOG-NTX-13, etc) and will enter the data into an electronic device/tablet according to the patient's response.~~

Treatment Visits 7, 8, 9, 10, 11, 12 (Weeks 12, 14, 16, 18, 20, 22 ±2 days)

Assessments at these visits are similar to Treatment Visit 1 with the following exceptions:

- Biochemistry parameters are assessed every 4 weeks i.e. Treatment Visit 7, 9 and 11 only **(samples to be taken in advance of the visit to have the results available before the start of the new chemotherapy and IMP cycle)**
- Serum β-HCG pregnancy test in ~~females of child-bearing potential~~ **WOCBP** will be performed every 4 weeks i.e. Treatment Visit 7, 9 and 11 only **(test to be taken in advance of the visit to have the results available before the start of the new chemotherapy and IMP cycle)**
- Blood Mn, and QoL/health status (EQ-5D-5L) are assessed at Treatment Visits 8 and 12 only.
- The ~~cold sensitivity patient~~ **Cold Sensitivity questionnaire (paper** diary**)** is completed before IMP infusion at Treatment Visit 8 and on days 1, 2, and 3 after the IMP infusion
- Health economic impact data are not collected at these visits

[…]

EOT Visit

The following assessments will be performed at ~~the end~~ **Day 14 (±3 days)** of the last cycle of mFOLFOX6:

- CEA
- Concomitant medications
- Physical examination
- Vital signs **and weight**
- ECOG performance status

- **Serum β-HCG pregnancy test in WOCBP**
- Hematology sample
- Biochemistry sample
- Blood Mn sample
- FACT/GOG-NTX-13
- Grooved Pegboard
- Vibration sensitivity test, performed using a graduated tuning fork
- EQ-5D-5L
- Pain NRS questionnaire
- AEs

All AEs including SAEs will be collected up to 30 days after the last administration of IMP and followed until resolution.

Reason:

To update for correct performance of hematology, biochemistry, serum pregnancy tests and vital signs, to harmonize wording regarding the Cold Sensitivity questionnaire and to correct study procedures regarding patient reported data collection.

The following amendment applies to:

Section no.	**Title**
Section 5.2.1.4	**Follow-up Phase**

Protocol text:

All patients will be followed for 24 months from first dose of IMP, with Assessment Visits every 3 months until Month 12 and every 6 months thereafter. Any patient who discontinued from IMP for any reason other than death or withdrawal of consent will continue to have CIPN and tumor assessments every 3 months.

This will be replaced by:

All patients will be followed for 24 months from first dose of IMP, with Assessment Visits every 3 months until Month 12 and every 6 months thereafter. Any patient who discontinued from IMP for any reason **during the treatment phase** other than death or withdrawal of consent will continue to have CIPN and tumor assessments every 3 months.

Reason:

To clarify follow-up procedures.

The following amendment applies to:

Section no.	**Title**
Section 5.3.2	**Blinding**

Protocol text:

This is a double-blind, placebo-controlled study. Blinding is critical to the integrity of this clinical study.

The IMP will be prepared by a qualified unblinded pharmacist/designee not otherwise involved in the study to ensure the blinding. The unblinded pharmacist/designee will prepare the IMP in an amber colored syringe masking the colored appearance on the IMP. The prepared IMP, i.e. the syringe and amber colored tubing, is provided to the blinded nurse who will perform the administration, not knowing which treatment is given.

To further protect the integrity of the study, no attempt should be made to measure Mn levels except from what is stated in the protocol.

This will be replaced by:

This is a double-blind, placebo-controlled study. Blinding is critical to the integrity of this clinical study.

The IMP will be prepared by a qualified unblinded pharmacist/designee not otherwise involved in the study to ensure the blinding. The unblinded pharmacist/designee will prepare the IMP in an amber colored syringe masking the colored appearance on the IMP. The prepared IMP, i.e. the syringe and amber colored tubing, is provided to the blinded nurse who will perform the administration, not knowing which treatment is given.

~~To further protect the integrity of the study, no attempt should be made to measure Mn levels except from what is stated in the protocol.~~

Reason:

To clarify blinding procedure requirements.

The following amendment applies to:

Section no.	**Title**
Section 5.4.2	**Packaging**

Protocol text:

PledOx, in colorless glass vials, each containing 20 mL of solution, will be packed into a kit. Kits will be transported frozen within -25°C to -15°C.

Placebo vials, each containing 20 mL of solution, will be packed into a kit. Kits will be transported frozen within -25°C to -15°C.

Temperature will be controlled during shipment.

This will be replaced by:

PledOx, in colorless glass vials, each containing 20 mL of solution, will be packed into a kit. **Each kit will contain 4 vials.** Kits will be transported frozen within -25°C to -15°C.

Placebo vials, each containing 20 mL of solution, will be packed into a kit. **Each kit will contain 4 vials.** Kits will be transported frozen within -25°C to -15°C.

~~Temperature will be controlled during shipment.~~

Reason:

To complete packaging information.

The following amendment applies to:

Section no.	**Title**
Section 5.4.3	**Labelling**

Protocol text:

The glass vials will be labelled in local language with separate labels for PledOx and for placebo. A booklet label in the local language and in compliance with applicable regulations and guidelines will be attached to the kit.

Each vial label will have a detachable part containing kit and batch number. This part will be removed by the unblinded pharmacist/designee preparing the correct dose and stuck onto the unblinded drug accountability log.

This will be replaced by:

~~The glass vials will be labelled in local language with separate labels for PledOx and for placebo.~~ A booklet label in the local language and in compliance with applicable regulations and guidelines will be attached to the **vials and the** kit.

Each vial label will have a detachable part containing ~~kit~~ **vial** and batch number. This part will be removed by the unblinded pharmacist/designee preparing the correct dose and stuck onto the unblinded drug accountability log.

Reason:

To correct labelling information.

The following amendment applies to:

Section no.	**Title**
Section 5.4.5	**Destruction of Surplus Medication**

Protocol text:

5.4.5 Destruction of Surplus Medication

This will be replaced by:

5.4.5 Destruction of ~~Surplus Medication~~ **IMP**

Reason:

To adjust heading for clarity.

The following amendment applies to:

Section no.	**Title**
Section 5.7	**Investigational Product Accountability**

Protocol text:

[…]

- All unused IMP and empty containers are stored until they have been checked by the monitor
- It is possible to reconcile records of all used and unused stocks as confirmed by Investigator's signature

This will be replaced by:

- All unused IMP and empty ~~containers~~ **vials** are stored until they have been checked by the monitor
- It is possible to reconcile records of all used and unused stocks as confirmed by Investigator's signature

Reason:

To correct accountability information.

The following amendment applies to:

Section no. **Title**

Section 5.8 **Code Breaks**

Protocol text:

Unblinding will be controlled through the randomization system. In case of such emergency that it is crucial for the Investigator (or any other treating physician) to know whether the patient has received PledOx or placebo. Unblinding can be requested in the system. If unblinding occurs, this must be documented in the patient's hospital records with date and name of the Investigator who decided to unblind the study treatment. The Sponsor must be notified prior to unblinding, except in cases where an immediate need for unblinding exists for reasons of patient safety at which time the Investigator can unblind the patient prior to notifying the Sponsor. However, the Investigator will have the final decision and unilateral right for unblinding in matters involving the safety of study patients.

[…]

The DSMB will monitor data in a blinded fashion during the study. Recalculation of sample size will also be performed blinded.

This will be replaced by:

Unblinding will be controlled through the randomization system. In case of such emergency that it is crucial for the Investigator (or any other treating physician) to know whether the patient has received PledOx or placebo~~. Unblinding~~ **, unblinding** can be requested in the system. If unblinding occurs, this must be documented in the patient's hospital records with date and name of the Investigator who decided to unblind the study treatment. The Sponsor must be notified prior to unblinding, except in cases where an immediate need for unblinding exists for reasons of patient safety at which time the Investigator can unblind the patient prior to notifying the Sponsor. However, the Investigator will have the final decision and unilateral right for unblinding in matters involving the safety of study patients.

[…]

The DSMB will monitor data ~~in a blinded fashion during the study. Recalculation of sample size will also be performed blinded~~ **according to the DSMB Charter**.

Reason:

To correct code break and Data Safety Monitoring Board procedures.

The following amendment applies to:

Section no. **Title**

Section 6.1 **Patient Inclusion criteria**

Protocol text:

[…]

4. The patient has undergone curative (R0) surgical resection performed not more than 8-12 weeks prior to randomization.

5. The patient has a postsurgical CEA level ≤1.5x upper limit of normal (ULN, in current smokers, CEA level ≤2.0 x ULN is allowed).

6. No prior anti-cancer therapy for CRC except radiotherapy or concomitant chemo-radiotherapy using a fluoropyrimidine alone for locoregional rectal cancer.

7. Patient indicated for up to 6 months of oxaliplatin-based chemotherapy and without any clinically observed neurological disorders.

[…]

This will be replaced by:

[…]

4. The patient has undergone curative (R0) surgical resection performed ~~not more than 8~~ **within** 12 weeks prior to randomization.

5. The patient has a postsurgical CEA level ≤1.5x upper limit of normal (ULN, in current smokers, CEA level ≤2.0 x ULN is allowed).

6. No prior anti-cancer therapy for CRC except radiotherapy or concomitant chemo-radiotherapy using a fluoropyrimidine alone for locoregional rectal cancer.

7. Patient indicated for up to 6 months of oxaliplatin-based chemotherapy and without ~~any clinically observed neurological disorders~~ **pathological findings of a neurologic exam performed prior to oxaliplatin treatment according to local practice.**

[…]

Reason:

To clarify inclusion criteria following FDA input to study PP06498 (Polar M) and to correct time frame for tumor surgery.

The following amendment applies to:

Section no.	**Title**
Section 6.2	**Patient Exclusion Criteria**

Protocol text:

[…]

4. Any prior oxaliplatin-based chemotherapy <1 year before the randomization.

[…]

This will be replaced by:

[…]

~~4. Any prior oxaliplatin-based chemotherapy <1 year before the randomization.~~

Reason:

To clarify exclusion criteria following FDA input to study PP06498 (Polar M).

The following amendment applies to:

Section no.	**Title**
Section 6.3.1	**Removal of Patients from Therapy or Assessment**

Protocol text:

The patient will be advised in the Informed Consent Form (ICF) that they have the right to withdraw from the study at any time without affecting their right to an appropriate follow-up investigation.

[…]

This will be replaced by:

The patient will be advised in the Informed Consent Form (ICF) that they have the right to withdraw from the study at any time without **providing an explanation and without** affecting their right to an appropriate follow-up investigation.

[…]

Reason:

To clarify patient rights.

The following amendment applies to:

Section no.	**Title**
Section 6.3.2	**Pregnancy**

Protocol text:

[…]

• Combined (estrogen and progestogen containing) hormonal contraception associated with inhibition of ovulation o Oral o Intravaginal o Transdermal
• Progestogen only hormonal contraception associated with inhibition of ovulation o Oral o Injectable o Implantable[2]
Highly Effective Methods That Are User Independent [1]
• Implantable progestogen only hormonal contraception associated with inhibition of ovulation • Intrauterine device (IUD) • Intrauterine hormone-releasing system (IUS) • Bilateral tubal occlusion
• **Vasectomized partner** A vasectomized partner is a highly effective contraception method provided that the partner is the sole male sexual partner of the WOCBP and the absence of sperm has been confirmed. If not, an additional highly effective method of contraception should be used.
• **Sexual abstinence** Sexual abstinence is considered a highly effective method only if defined as refraining from heterosexual intercourse during the entire period of risk associated with the study treatment. The reliability of sexual abstinence needs to be evaluated in relation to the duration of the study and the preferred and usual lifestyle of the participant.

[1]: Typical use failure rates may differ from those when used consistently and correctly. Use should be consistent with local regulations regarding the use of contraceptive methods for participants participating in clinical studies.
[2]: Contraception methods that in the context of this guidance are considered to have low user dependency

<u>This will be replaced by:</u>

[…]

Highly Effective Methods That Are User Independent[1]
• Combined (estrogen and progestogen containing) hormonal contraception associated with inhibition of ovulation o Oral o Intravaginal o Transdermal
• Progestogen only hormonal contraception associated with inhibition of ovulation o Oral o Injectable o Implantable[2]
~~Highly Effective Methods That Are User Independent~~ [1]
• ~~Implantable progestogen only hormonal contraception associated with inhibition of ovulation~~ • Intrauterine device (IUD)[2] • Intrauterine hormone-releasing system (IUS)[2] • Bilateral tubal occlusion[2]
• Vasectomized partner[2]

A vasectomized partner is a highly effective contraception method provided that the partner is the sole male sexual partner of the ~~female of child bearing potential~~ **WOCBP** and the absence of sperm has been confirmed. If not, an additional highly effective method of contraception should be used.
• Sexual abstinence Sexual abstinence is considered a highly effective method only if defined as refraining from heterosexual intercourse during the entire period of risk associated with the study treatment. The reliability of sexual abstinence needs to be evaluated in relation to the duration of the study and the preferred and usual lifestyle of the participant.

[1]: Typical use failure rates may differ from those when used consistently and correctly. Use should be consistent with local regulations regarding the use of contraceptive methods for participants participating in clinical studies.
[2]: Contraception methods that in the context of this guidance are considered to have low user dependency

Reason:

To clarify highly effective contraception methods.

The following amendment applies to:

Section no. **Title**

Section 7.2 **Method of Assigning Patients to Treatment Groups**

Protocol text:

[…]

Randomization is performed centrally.

This will be replaced by:

[…]

Randomization is performed centrally **using an Interactive Web Response System**.

Reason:

To clarify randomization procedures.

The following amendment applies to:

Section no. **Title**

Section 7.4.1 **PledOx/Placebo Administration**

Protocol text:

PledOx/Placebo will be administered by an i.v. infusion over approximately 5 minutes, 10 minutes prior to each mFOLFOX6.

[…]

* A specific template for calculation will be provided all patients will receive 22 mL i.e. the volume of PledOx/placebo will be based on body weight

[…]

This will be replaced by:

PledOx/Placebo will be administered by an i.v. infusion over approximately 5 minutes, 10 minutes prior to each mFOLFOX6 **cycle**.

[…]

* A specific template for **dose** calculation will be provided all patients will receive **20** ~~22~~ mL i.e. the volume of PledOx/placebo will be based on body weight

[…]

Reason:

To correct dose volume.

The following amendment applies to:

Section no. **Title**

Section 7.4.2 **Chemotherapy (mFOLFOX6) Administration**

Protocol text:

All patients will receive 5-FU, calcium-folinate or calcium-levofolinate and oxaliplatin, according to the mFOLFOX6 regimen, and pre-treatment with anti-emetics and corticosteroids will be given, as specified in Table 2 and the approved SmPC. The infusion line and the port should be flushed between the administrations. This is particularly important between IMP and oxaliplatin administration.

Table 2 Chemotherapy Administration Schedule

Drug	Dose	Administration	Duration	Infusate/ Preparation	Time point
Ondansetron*	8 mg*	i.v. injection			At least 30 min prior to start of chemotherapy
Beta-methasone*	8 mg*	i.v. injection			At least 30 min prior to start of chemotherapy
Oxaliplatin	85 mg/m^2	i.v. infusion	0-2 hours	0.5 L glucose 5%	Start of chemotherapy (regarded as time point 0)
Ca-levofolinate or Ca-folinate	100 mg/m^2 200 mg/m^2	i.v. infusion i.v. infusion	0-2 hours 0-2 hours	0.5 L glucose 5% 0.5 L glucose 5%	Start of chemotherapy (regarded as time point 0)
5-FU	400 mg/m^2	i.v. bolus		Chase infusion 0.5 L NaCl 0.9%	Approximately 2 h after time point 0
5-FU	2400 mg/m^2	i.v. continued infusion	2-48 hours	Chase infusion	Approximately 2 h after time point 0

This will be replaced by:

All patients will receive 5-FU, calcium-folinate or calcium-levofolinate and oxaliplatin, according to the mFOLFOX6 regimen, and pre-treatment with anti-emetics and corticosteroids will be given, as specified in Table 2 and the approved SmPC**, approved label and national guidance**. The infusion line and the port should be flushed between the administrations. This is particularly important between IMP and oxaliplatin administration.

Table 2 Chemotherapy Administration Schedule

Drug	Dose	Administration	Duration	Infusate/ Preparation	Time point
Ondansetron*	8 mg*	i.v. injection			At least 30 min prior to start of chemotherapy
Beta-methasone*	8 mg*	i.v. injection			At least 30 min prior to start of chemotherapy
Oxaliplatin	85 mg/m^2	i.v. infusion	0-2 hours	0.5 L glucose 5%	Start of chemotherapy (regarded as time point 0)
Ca-levofolinate or Ca-folinate	100 ~~mg/m^2~~ **or** 200 mg/m^2 **200 or 400 mg/m^2**	i.v. infusion i.v. infusion	0-2 hours 0-2 hours	0.5 L glucose 5% 0.5 L glucose 5%	Start of chemotherapy (regarded as time point 0)
5-FU	400 mg/m^2	i.v. bolus		~~Chase infusion 0.5 L~~ **Recommendation 50-100 ml** NaCl 0.9%	Approximately 2 h after time point 0
5-FU	2400 mg/m^2	i.v. continued infusion	2-48 hours	~~Chase infusion~~	Approximately 2 h after time point 0

Reason:

To clarify doses of chemotherapy.

The following amendment applies to:

Section no. **Title**

Section 7.5 **Dose Adjustment Criteria**

Protocol text:

Any change to the IMP (i.e. PledOx or placebo), chemotherapy (mFOLFOX6), or other therapy (e.g. biologics) must be recorded in the eCRF, including the reason for the change.

This will be replaced by:

Any **dose** change to the IMP (i.e. PledOx or placebo), chemotherapy (mFOLFOX6), or other therapy (e.g. biologics) must be recorded in the eCRF, including the reason for the change.

Reason:

To clarify doses adjustment criteria for the IMP.

The following amendment applies to:

Section no. **Title**

Section 7.5.1 **Treatment Requirements and Delays**

Protocol text:

Prior to each administration of chemotherapy, the patient's hematology must be adequate (as per local practice), and all toxicities (other than alopecia) must have resolved to Grade ≤2 (with the exception of platelets i.e. Grade <1). In case of febrile neutropenia, Table 3 treatment will be discontinued. All pre-treatment lab data including hematology, biochemistry, pregnancy testing, etc. may not be older than 72 hours before Day 1 of applicable cycle.

If these criteria are not met at the time of a planned treatment, the following general rules for the management of treatment delays apply:

- In the case of mFOLFOX6 related toxicity (see Section 7.5.2), the start of the next cycle will be delayed until recovery. In addition, PledOx/placebo treatment should be withheld until recovery, and then restarted in parallel with mFOLFOX6.

If administration of mFOLFOX6 is delayed for more than 6 weeks (3 cycles) after the first day of the previous cycle, treatment with that therapy will be discontinued.

This will be replaced by:

Prior to each administration of chemotherapy, the patient's hematology must be adequate (as per local practice), and all toxicities (other than alopecia) must have resolved to Grade ~~≤2 (with the exception of platelets i.e. Grade~~ **<0 or 1**~~)~~.

In case of febrile neutropenia, **and the patient's recovery to Grade 0 or 1, please see Table 5** ~~3 treatment will be discontinued~~ **for dose reduction recommendations**. All pre-treatment lab data including hematology, biochemistry, pregnancy testing, etc. may not be older than 72 hours before Day 1 of applicable cycle.

If these criteria are not met at the time of a planned treatment, the following general rules for the management of treatment delays apply:

- In the case of mFOLFOX6 related toxicity (see Section 7.5.2), the start of the next cycle will be delayed until ~~recovery.~~ **toxicities (other than alopecia) are resolved to Grade 0 or 1.** In addition, PledOx/placebo treatment should be withheld until recovery, and then restarted in parallel with mFOLFOX6.

If administration of mFOLFOX6 is delayed for more than 6 weeks (3 cycles) after the first day of the previous cycle, treatment with that therapy will be discontinued.

Reason:

To update procedure for treatment delays following Coordinating Investigator input.

The following amendment applies to:

Section no. **Title**

Section 7.5.2 **Chemotherapy Related AEs and Dose Modification**

Protocol text:

The following sections include guidelines for dose modifications and delays associated with specific symptoms.

Dose limiting toxicities (DLTs) for hematological and neurological toxicities and diarrhea are listed below. However, every Grade 3 or 4 toxicities according to NCI-CTCAE v4.03 that, in the opinion of the Investigator, put the patient at risk should be evaluated and adequate dose reduction/treatment delay should be considered. All dose modifications and the reasons should be recorded in applicable sections of the eCRF.

If other symptoms not discussed in this protocol occur, the Investigator should be consulted regarding the appropriate course of action with regard to dose modification or delay.

Refer to Section 9.1.1.5 for chemotherapy related AEs.

This will be replaced by:

The following sections include guidelines for dose modifications and delays associated with specific symptoms.

~~Dose limiting toxicities (DLTs) for hematological and neurological toxicities and diarrhea are listed below. However, every~~ **Every** Grade 3 or 4 toxicity according to NCI-CTCAE v4.03 that, in the opinion of the Investigator, put the patient at risk should be evaluated and adequate dose reduction/treatment delay should be considered. All dose modifications and the reasons should be recorded in applicable sections of the eCRF.

If other symptoms not discussed in this protocol occur, the **treating** Investigator should be consulted regarding the appropriate course of action with regard to dose modification or delay.

~~Refer to Section 9.1.1.5 for chemotherapy related AEs.~~

Reason:

To clarify chemotherapy related dose modifications.

The following amendment applies to:

Section no. **Title**

Section 7.5.2.1 **Dose Modification of Oxaliplatin due to Neuropathy**

Protocol text:

[...]

Modification of infusion duration of oxaliplatin; increased of oxaliplatin infusion time from 1-2 hours to 6 hours in case of NCI-CTC Grade ≤2 allergy or in case of laryngeal spasm syndrome.

If treatment with oxaliplatin must be stopped due to toxicity, treatment with 5-FU/folinate and IMP may be continued.

Table 1 Definition of Neuropathy According to NCI-CTCAE v4.03

Grade	Definition
Grade 0	No change or none
Grade 1	Mild paresthesia, loss of deep tendon reflexes
Grade 2	Mild or moderate objective sensory loss, moderate paresthesia
Grade 3	Severe objective sensory loss or paresthesia
Grade 4	Not applicable

This will be replaced by:

[…]

Modification of infusion duration of oxaliplatin**:** ~~; increased of oxaliplatin~~ infusion time **may be increased** from 1-2 hours to 6 hours in case of NCI-~~CTC~~**CTCAE v4.03** Grade ≤2 allergy or in case of laryngeal spasm syndrome.

If treatment with oxaliplatin must be stopped due to toxicity, treatment with 5-FU/folinate and IMP may be continued.

Table 2 Definition of Neuropathy According to NCI-CTCAE v4.03

Grade	Definition**: Peripheral sensory neuropathy**
Grade 0	No change or none
Grade 1	~~Mild paresthesia~~ **Asymptomatic**; loss of deep tendon reflexes **or paresthesia**
Grade 2	~~Mild or moderate objective sensory loss, moderate paresthesia~~ **Moderate symptoms, limiting instrumental ADL**
Grade 3	Severe ~~objective sensory loss or paresthesia~~ **symptoms; limiting self-care ADL**
Grade 4	~~Not applicable~~ **Life-threatening consequences; urgent intervention indicated**
Grade 5	**Death**

ADL = activities of daily living

Reason:

To update and harmonize dose modification of oxaliplatin due to neuropathy.

The following amendment applies to:

Section no. Title

Section 7.5.2.2 Central Nervous System Toxicity

Protocol text:

All central neurologic toxicity should be evaluated by the Investigator and in case of Grade 3 and 4 toxicities, according to NCI-CTCAE v4.03, mFOLFOX6 treatment should be stopped until neurological symptoms is reduced to less than Grade 2.

This will be replaced by:

All central neurologic toxicity should be evaluated by the Investigator and in case of Grade 3 and 4 toxicities, according to NCI-CTCAE v4.03, mFOLFOX6 treatment should be stopped until neurological symptoms is reduced to ~~less than~~ Grade ~~2~~ **1 or 0**.

Reason:

To harmonize information on toxicity.

The following amendment applies to:

Section no. **Title**

Section 7.5.2.3 **Hematological Dose Modification of Oxaliplatin and 5-FU**

Protocol text:

Hematological dose modifications should be performed according to the table below (Table 5), but may also be performed according to the local practice.

Table 3 Hematological Dose Modification of Oxaliplatin and 5-FU

Neutropenia	**Duration (counting from pre-dose of the cycle)**	**Action**
Grade 3/4 neutropenia first time	≤7 days	No dose reduction when recovered
Grade 3/4 neutropenia second time	≤7 days	Dose reduction to 75% of full dose when recovered (which can mean dose delay)
Grade 3/4 neutropenia first time	>7 days	Dose reduction to 75% of full dose when recovered (which can mean dose delay)
Grade 4 neutropenia second time	>7 days	Discontinue treatment
Febrile neutropenia	-	Discontinue treatment
Platelet **count decreased** ~~Count (x10^9/L)~~	Oxaliplatin	5-FU
>100	No dose reduction	No dose reduction
99-75	Dose reduction 75%	Dose reduction 75%
74-50	Dose reduction 50%	Dose reduction 50%
<50	Delay treatment 1 week	

Definition for neutropenia according to NCI-CTCAE v4.03 based on ANC:

- Grade 1: < Lower limit of normal and > 1.5 x 10^9/L (> 1500/mm^3)
- Grade 2: < 1.5 x 10^9/L (< 1500/mm^3) and > 1.0 x 10^9/L (> 1000/mm^3)
- Grade 3: < 1.0 x 10^9/L (< 1000/mm^3) and > 0.5 x 10^9/L (> 500/mm^3)
- Grade 4: < 0.5 x 10^9/L (< 500/mm^3)
- Grade 5: Death

This will be replaced by:

Prior to each administration of chemotherapy, the patient's hematology must be adequate (as per local practice), and toxicities must have resolved to Grade 0 or 1.

Hematological dose modifications **(based on nadir)** should be performed according to the table below (Table 5), but may also be performed according to the local practice.

Table 5 Hematological Dose Modification of Oxaliplatin and 5-FU

Neutropenia	Duration (counting from pre-dose of the cycle)	Action
Grade 3/4 ~~neutropenia~~ first time	≤7 days	No dose reduction when recovered
Grade 3/4 ~~neutropenia~~ second time	≤7 days	Dose reduction to 75% of full dose when recovered (which can mean dose delay)
Grade 3/4 ~~neutropenia~~ first time	>7 days	Dose reduction to 75% of full dose when recovered (which can mean dose delay)
Grade **3**/4 ~~neutropenia~~ second time	>7 days	~~Discontinue treatment~~ **Dose reduction to 75% of previous dose when recovered (which can mean dose delay)**
Febrile neutropenia	-	~~Discontinue treatment~~ **Dose reduction to 75% of previous dose when recovered (which can mean dose delay)**
Platelet count ~~(x10^9/L)~~ **decreased**	**Oxaliplatin**	**5-FU**
~~>100~~ **Grade 1/2**	No dose reduction	No dose reduction
~~99-75~~ **Grade 3/4**	Dose reduction 75%	Dose reduction 75%
~~74-50~~	~~Dose reduction 50%~~	~~Dose reduction 50%~~
~~<50~~	~~Delay treatment 1 week~~	

Definition for neutropenia according to NCI-CTCAE v4.03 based on ANC:

- Grade 1: <Lower limit of normal ~~and >~~ - 1.5 x 10^9/L ~~(>~~ **(**<1500/mm^3)
- Grade 2: <1.5 x 10^9/L (<1500/mm^3) and ~~>~~ ≥1.0 x 10^9/L ~~(>~~ **(**≥1000/mm^3)
- Grade 3: <1.0 x 10^9/L (<1000/mm^3) and ~~>~~ ≥0.5 x 10^9/L ~~(>~~ **(**≥500/mm^3)
- Grade 4: <0.5 x 10^9/L (<500/mm^3)
- ~~Grade 5: Death~~

Definition of platelet count decreased according to NCI-CTCAE v4.03

- **Grade 1: <Lower limit of normal -75.0 x 10^9/L (<75.000/mm^3)**
- **Grade 2:** <75**.0 x 10^9/L (<75,000/mm^3) and ≥50.0 x 10^9/L (≥50,000/mm^3)**
- **Grade 3: <50.0 x 10^9/L (<50,000/mm^3) and ≥25.0 x 10^9/L (≥25,000/mm^3)**
- **Grade 4: <25.0 x 10^9/L (<25,000/mm^3)**

Reason:

To clarify and harmonize dose modification procedures of oxaliplatin and 5-FU based on hematology.

The following amendment applies to:

Section no.	**Title**
Section 7.6	**Concomitant Medication**

Protocol text:

[…]

Granulate-macrophage colony stimulating factor (GM-CSF) and granulocyte colony stimulating factor (G-CSF) is not allowed as prophylactic treatment, but may be used to treat side effects at the Investigator's discretion.

[…]

This will be replaced by:

[…]

~~Granulate-macrophage~~ **The use of growth factors (i.e. granulocyte-**colony stimulating factor ~~(~~**[G-CSF]) in patients who require G-CSF or** GM-CSF~~) and granulocyte colony stimulating factor (G-CSF) is not~~ **therapeutically will be** allowed ~~as prophylactic treatment, but may~~ **at the discretion of the Investigator and may also** be used to treat side effects at the Investigator's discretion.

[…]

Reason:

To clarify use of growth factors as concomitant medication.

The following amendment applies to:

Section no.	**Title**
Section 8.1.1.1	**Neuropathy Assessment - FACT/GOG-NTX**

Protocol text:

[…]

Patients will complete this questionnaire electronically (on a hand-held device/tablet) in the hospital, before IMP infusion (Day 1 of each mFOLFOX6 cycle) during the Treatment Phase and at each Assessment Visit.

Note: If a patient does not attend the hospital for an Assessment Visit, the study staff will conduct a recorded telephone call with the patient to collect the relevant data. In such cases, study staff will read through the questionnaires (e.g. FACT/GOG-NTX-13, etc) and will enter the data into an electronic device/tablet according to the patient's response.

This will be replaced by:

[…]

will complete this questionnaire electronically (on a hand-held device/tablet) in the hospital, before IMP infusion (Day 1 of each mFOLFOX6 cycle) during the Treatment Phase and at each Assessment Visit.

~~Note: If a patient does not attend the hospital for an Assessment Visit, the study staff will conduct a recorded telephone call with the patient to collect the relevant data. In such cases,~~

~~study staff will read through the questionnaires (e.g. FACT/GOG-NTX-13, etc) and will enter the data into an electronic device/tablet according to the patient's response.~~

Reason:

To correct study procedures regarding patient reported data collection.

The following amendment applies to:

Section no.	**Title**
Section 9.1.1.1	**Adverse Event Definition**

Protocol text:

An AE in this study is defined as any untoward medical occurrence in a patient who has received IMP. The occurrence does not necessarily need to have a causal relationship with the IMP. An AE can therefore be any unfavorable and unintended sign (including an abnormal laboratory finding), symptom, or disease temporally associated with the administration of IMP, whether or not causally related.

The occurrence of an AE may come to the attention of study personnel during study visits and interviews of a study recipient presenting for medical care, or upon review by a study monitor who is scrutinizing relevant source data.

This will be replaced by:

An AE in this study is defined as any untoward medical occurrence in a patient who has received IMP. The occurrence does not necessarily need to have a causal relationship with the IMP. An AE can therefore be any unfavorable and unintended sign (including an abnormal laboratory finding), symptom, or disease temporally associated with the administration of IMP, whether or not causally related.

The occurrence of an AE may come to the attention of study personnel during study visits and interviews of a study recipient presenting for medical care, or upon review by a study monitor who is scrutinizing relevant source data.

All AEs, including chemotherapy related toxicities, will be graded using the NCI-CTCAE v4.03.

Reason:

To update the definition and grading of adverse events following FDA input to study PP06498 (Polar M).

The following amendment applies to:

Section no.	**Title**
Section 9.1.1.9	**Recording and Reporting Adverse Event/Intercurrent Illnesses**

Protocol text:

[…]

Progression of disease should not be reported as an SAE.

Deaths due solely due to progression of disease should not be reported as an SAE.

[…]

This will be replaced by:

[…]

Progression of disease should not be reported as an SAE.

~~Deaths due solely due to progression of disease should not be reported as an SAE.~~

[…]

Reason:

To correct the reporting procedure for deaths due to progressive disease.

The following amendment applies to:

Section no.	**Title**
Section 9.1.2	**Laboratory Evaluation**

Protocol text:

[…]

Laboratory values outside the normal ranges will be flagged in the laboratory report. For each flagged value the Investigator should mark the value as “Clinically Significant” (CS) or “Not Clinically Significant” (NCS).

This will be replaced by:

~~Laboratory values~~ **For each value** outside the normal ranges will ~~be flagged in the laboratory report. For each flagged value,~~ the Investigator should mark the value as “Clinically Significant” (CS) or “Not Clinically Significant” (NCS).

Reason:

To correct procedure for evaluation of laboratory values outside normal ranges.

The following amendment applies to:

Section no.	**Title**
Section 9.1.2.2	**Biochemistry**

Protocol text:

[…]

Biochemistry: Alkaline phosphatase (ALP), total bilirubin, ALT, AST, serum creatinine, and albumin.

This will be replaced by:

[…]

Biochemistry: Alkaline phosphatase (ALP), total bilirubin, ALT, AST, serum creatinine, and albumin.

Local laboratories will be used to analyze biochemistry samples.

Reason:

To harmonize laboratory sample analysis procedures.

The following amendment applies to:

Section no. **Title**

Section 9.1.2.3 **Blood Manganese**

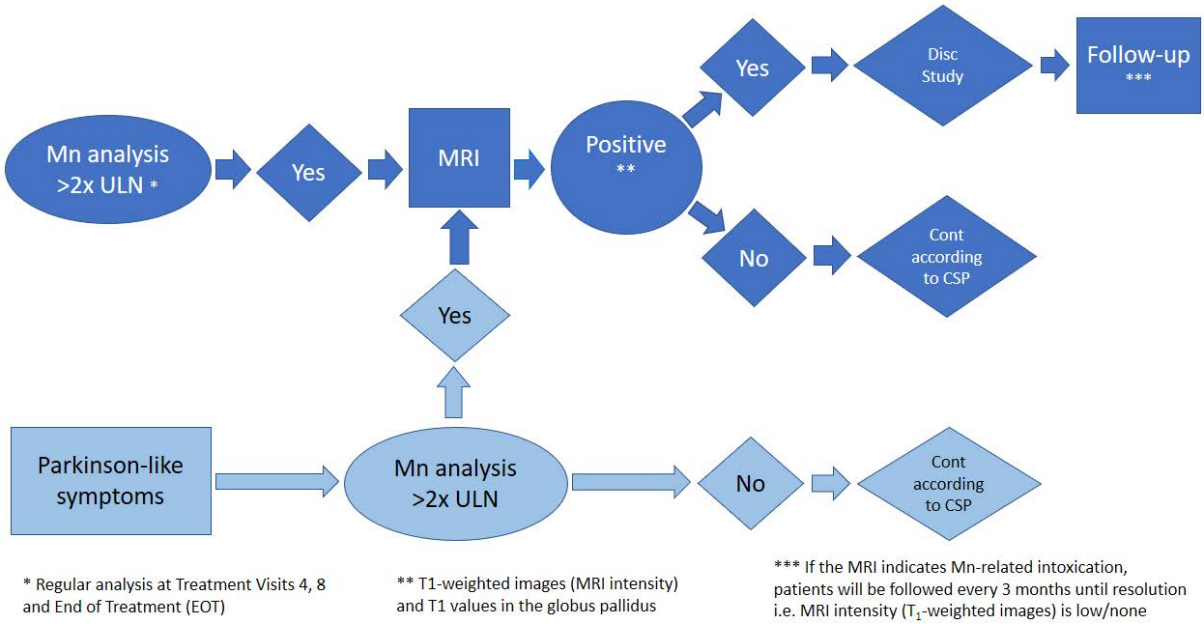
Protocol text:

[…]

Figure 2 shows a flow chart of monitoring increased Mn levels and/or Parkinson-like symptoms. Patients that show Mn concentrations >2 x ULN before Cycle 5 or Cycle 9 will be referred to a brain MRI investigation and a neurological examination (see Section 9.1.3.6). IMP treatment should be interrupted until MRI results are available. If the MRI is positive (i.e. T1-weighted images ([MRI intensity)] and T1 images in the globus pallidus), the patient will be discontinued from the study and followed every 3 months until resolution (i.e. MRI intensity is low/none [Tagaki et al. 2002]).

[…]

Figure 2 Flow Chart of Monitoring Increased Mn Levels and/or Parkinson-like Symptoms



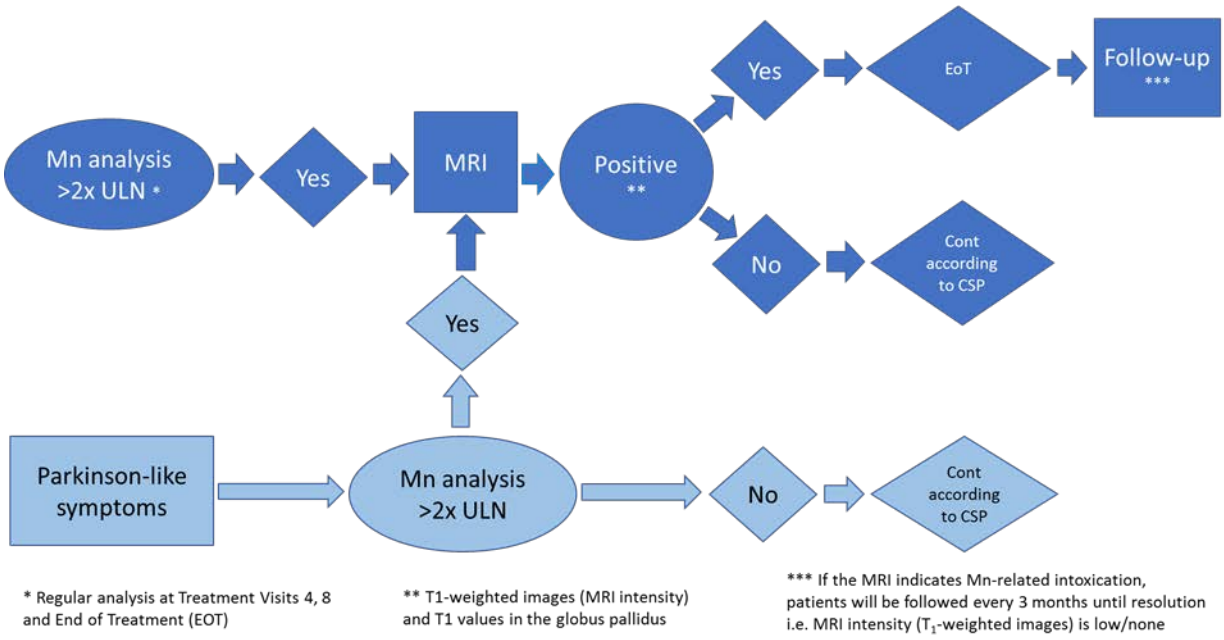
This will be replaced by:

[…]

Figure 2 shows a flow chart of monitoring increased Mn levels and/or Parkinson-like symptoms. Patients that show Mn concentrations >2 x ULN before Cycle 5 or Cycle 9 will be referred to a brain MRI investigation and a neurological examination (see Section 9.1.3.6). IMP treatment ~~should be interrupted~~ **continue** until MRI results are available. If the MRI is positive (i.e. T1-weighted images ([MRI intensity)] and T1 images in the globus pallidus), the patient will be discontinued from the study and followed every 3 months until resolution (i.e. MRI intensity is low/none [Tagaki et al. 2002]).

[…]

Figure 2 Flow Chart of Monitoring Increased Mn Levels and/or Parkinson-like Symptoms



Reason:

To update and correct Mn analysis procedures.

The following amendment applies to:

Section no.	**Title**
Section 9.1.3.1	**Vital Signs**

Protocol text:

Vital signs will be recorded at time points according to the Schedule of Assessments (Appendix 1).

Vital signs include resting pulse (beats per minute), body temperature, and resting systolic and diastolic blood pressure and should be measured on the right arm after at least 5 minutes of supine rest.

Weight will be measured at Screening, each Treatment visit, and the EOT visit.

Height will be recorded at Screening only.

This will be replaced by:

Vital signs will be recorded at time points according to the Schedule of Assessments (Appendix 1).

Vital signs include resting pulse (beats per minute), body temperature, and resting systolic and diastolic blood pressure and should be measured on the ~~right~~ **same** arm after at least 5 minutes of supine rest.

Weight will be measured at Screening, each Treatment visit, and the EOT visit.

Height will be recorded at Screening only.

Reason:

To correct vital sign measurement method.

The following amendment applies to:

Section no.	**Title**
Section 9.1.3.2	**Physical Examination**

Protocol text:

[…]

The physical examination should be performed according to local clinical praxis. Any abnormal findings should be recorded and judged as CS or NCS by the Investigator. Any clinically significant abnormalities should be recorded as medical history if detected at screening or baseline, or as AEs if detected after the start of treatment.

This will be replaced by:

[…]

The physical examination **including a neurologic assessment (specifically including, but not limited to, tendon reflexes and sensation)** should be performed according to local clinical ~~praxis~~ **practice prior to oxaliplatin treatment**. Any abnormal**/pathological** findings should be recorded and judged as CS or NCS by the Investigator. Any clinically significant abnormalities should be recorded as medical history if detected at screening or baseline, or as AEs if detected after the start of treatment.

Reason:

To update physical examination procedures following FDA input to study PP06498 (Polar M).

The following amendment applies to:

Section no.	**Title**
Section 9.1.3.5	**Tumor/Disease Assessments**

Protocol text:

Baseline radiographic assessment of disease will be performed within 42 days prior to surgery or after surgery if not done previously. The period from surgery to start of therapy should be as soon as possible, preferably within 8 to 12 weeks. A CT or MRI scan of the thorax, abdomen and pelvis will be performed according to the Schedule of Assessments (Appendix 1) to identify any signs for CRC recurrence. CT or MRI should be in accordance with local guidelines and practice, but for each patient the same assessment should be performed throughout the study. CT/MRI scans will be read locally.

Assessment of DFS will include objective review of signs or symptoms of recurrence of CRC. Signs and symptoms may include any assessment result that, in the opinion of the Investigator, provides evidence for the recurrence of the disease, including CT or MRI, CEA, hematology, biochemistry, etc.

This will be replaced by:

Baseline radiographic assessment of disease will be performed within 42 days prior to surgery or after surgery if not done previously. The period from surgery to start of therapy should be as soon as possible, preferably within ~~8 to~~ 12 weeks. ~~A~~

The CT or MRI scan of the thorax, abdomen and pelvis will be performed according to the Schedule of Assessments (Appendix 1) to identify any signs for CRC recurrence. **The method chosen,** CT **(preferred)** or MRI**, is optional and** should be in accordance with local guidelines and practice, but for each patient the same assessment should be performed throughout the study. CT/MRI scans will be read locally.

Assessment of DFS will include objective review of signs or symptoms of recurrence of CRC. Signs and symptoms may include any assessment result that, in the opinion of the Investigator, provides evidence for the recurrence of the disease, including CT or MRI, CEA, hematology, biochemistry, etc.

Reason:

To clarify tumor assessment procedures.

The following amendment applies to:

Section no.	**Title**
Section 9.1.3.6	**MRI of the Central Nervous System and Neurological Examination**

Protocol text:

If blood Mn is found to be elevated (>2x ULN) during scheduled Mn samples, an MRI of the CNS and neurological examination will be performed to evaluate any cerebral neurological impact.

The following assessments may optionally be included in the neurological examination:

[…]

This will be replaced by:

If blood Mn is found to be elevated (>2x ULN) during scheduled Mn samples, an MRI of the CNS and neurological examination will be performed to evaluate any cerebral neurological impact.

The following assessments may ~~optionally~~ be included in the neurological examination:

[…]

Reason:

To clarify MRI and neurological examination procedures.

The following amendment applies to:

Section no.	**Title**
Section 9.2	**Appropriateness of Measurements**

Protocol text:

[…]

The maximum volume of blood collected per patient at a visit will be 20.5 mL. Details are shown in Table 8. Some of the assessments shown below will not be collected at all visits (see Appendix 1 for the Schedule of Assessments).

Table 8. Maximum volume of blood collected at a single visit

Assessment	Volume (mL)
CEA	0.6
Pregnancy test (serum) in females of child-bearing potential only	4.9
Hematology	4.5
Biochemistry	4.5
Mn	6.0
Total	20.5

This will be replaced by:

[…]

The maximum volume of blood collected **for study purposes** per patient at a visit will be 20.5 mL. Details are shown in Table 8. Some of the assessments shown below will not be collected at all visits (see Appendix 1 for the Schedule of Assessments).

Table 8. Maximum volume of blood collected **for study purposes** at a single visit

Assessment	Volume (mL)
CEA	0.6
Pregnancy test (serum) in ~~females of child-bearing potential~~ **WOCBP** only	4.9
Hematology	4.5
Biochemistry	4.5
Mn	6.0
Total	20.5

Reason:

To update text and table for clarity and consistency.

The following amendment applies to:

Section no. **Title**

Section 10.1.2 **General Considerations**

Protocol text:

Demographic and baseline characteristics (e.g. age [years], gender [male, female], race [White, Asian, Black, other], body weight [kg], height [cm], etc) will be summarized by treatment group and overall for All Randomized Patients and the FAS.

Baseline characteristics and endpoints will be presented using descriptive statistics: continuous variables in terms of number of patients (n), mean, median, standard deviation (SD), and range (minimum and maximum). Categorical variables will be summarized by counts and percentages. Unless otherwise stated the calculation of proportions will be based on the sample size of the population of interest.

[…]

This will be replaced by:

Demographic and baseline characteristics (e.g. age [years], gender [male, female], race [White, Asian, Black, other], body weight [kg], height [cm], etc) will be summarized by treatment group and overall ~~for All Randomized Patients and the FAS~~.

Baseline characteristics and endpoints will be presented using descriptive statistics: continuous variables in terms of number of patients (n), mean, median, standard deviation (SD), and range (minimum and maximum). Categorical variables will be summarized by counts and percentages. Unless otherwise stated the calculation of proportions will be based on the sample size of the population of interest.

Missing data will be handled according to the SAP.

[…]

Reason:

To clarify analysis procedures.

The following amendment applies to:

Section no.	**Title**
Section 10.1.4	**Primary Efficacy Endpoint Analysis**

Protocol text:

[…]

The OR, 95% CI and p-value using the Cochran-Mantel-Haenszel test for the difference in event rates between the 2 treatment groups will be presented. The results from the Cochran-Mantel-Haenszel test are considered the primary analysis results using the FAS.

Missing data will be handled using the approach described above

[…]

This will be replaced by:

[…]

The OR, 95% CI and p-value using the Cochran-Mantel-Haenszel test for the difference in event rates between the 2 treatment groups will be presented. The results from the Cochran-Mantel-Haenszel test are considered the primary analysis results using the FAS.

~~Missing data will be handled using the approach described above~~

[…]

Reason:

To update analysis procedures for consistency.

The following amendment applies to:

Section no.	**Title**
Section 10.1.7	**Safety Analysis**

Protocol text:

[…]

Laboratory results will be classified according to NCI-CTCAE v.4.03. Incidence of laboratory abnormalities will be summarized; laboratory results not corresponding to an NCI-CTCAE v.4.03 term will not be graded.

The results from physical examinations, vital signs measurement, and ECG evaluation will be tabulated. Descriptive statistics will be provided as appropriate.

This will be replaced by:

[…]

Laboratory results will be classified according to NCI-CTCAE v.4.03. Incidence of laboratory abnormalities will be summarized; laboratory results not corresponding to an NCI-CTCAE v.4.03 term will not be graded.

The results from physical examinations, **and** vital signs measurement~~, and ECG evaluation~~ will be tabulated. Descriptive statistics will be provided as appropriate.

Reason:

To update analysis procedures for consistency.

The following amendment applies to:

Section no.	**Title**
Section 10.3	**Level of Significance**

Protocol text:

[…]

The second family will include the key secondary endpoints (see below) and they will be tested using an overall type I error rate of 0.05 by means of a hierarchical procedure, i.e. in a predefined order to control for multiplicity within this family. This will be done only if the primary endpoint has met its criteria for rejecting the null hypothesis, i.e. if $p<0.05$ for PledOx 5 µmol/kg vs placebo. Subsequently, the test for the next secondary endpoints within that family will be performed only if the preceding hypothesis test shows $p<0.05$. The predefined (hierarchical) order to test the key secondary variables will be the following:

1. Proportion of patients (with mild, moderate or severe chronic CIPN) scoring 2, 3 or 4, in at least one of the first 4 items of the FACT/GOG-NTX-13 (i.e. FACT/GOG-NTX-4), targeting numbness, tingling or discomfort in hands and/or feet, 9 months after first dose of IMP.
2. Mean change from baseline in sensitivity to touching cold items on day 2 of Cycle 4 of mFOLFOX6 chemotherapy, as assessed by the Cold Sensitivity questionnaire.
3. Mean cumulative dose of oxaliplatin administered per patient during mFOLFOX6 chemotherapy, 9 months after the first dose of IMP.

4. Mean change from baseline in vibration sense, on the lateral malleolus (left and right), using a graduated tuning fork, at 9 months after the first dose of IMP.

This will be replaced by:

[…]

The second family will include the key secondary endpoints (see below) and they will be tested using an overall type I error rate of 0.05 by means of a hierarchical procedure, i.e. in a predefined order to control for multiplicity within this family. This will be done only if the primary endpoint has met its criteria for rejecting the null hypothesis, i.e. if $p<0.05$ for PledOx 5 µmol/kg vs placebo. Subsequently, the test for the next secondary endpoints within that family will be performed only if the preceding hypothesis test shows $p<0.05$. The predefined (hierarchical) order to test the key secondary variables will be the following:

- ~~1.~~ Proportion of patients (with mild, moderate or severe chronic CIPN) scoring 2, 3 or 4, in at least one of the first 4 items of the FACT/GOG-NTX-13 (i.e. FACT/GOG-NTX-4), targeting numbness, tingling or discomfort in hands and/or feet, 9 months after first dose of IMP.
- ~~2.~~ Mean change from baseline in sensitivity to touching cold items on day 2 of Cycle 4 of mFOLFOX6 chemotherapy, as assessed by the Cold Sensitivity questionnaire.
- ~~3.~~ Mean cumulative dose of oxaliplatin administered per patient during mFOLFOX6 chemotherapy, 9 months after the first dose of IMP.

~~4. Mean change from baseline in vibration sense, on the lateral malleolus (left and right), using a graduated tuning fork, at 9 months after the first dose of IMP.~~

Reason:

To update analysis procedures for consistency.

The following amendment applies to:

Section no.	**Title**
Appendix 1	**Schedule of Assessments**

Note: the substantial changes for the Schedule of Assessments included removal of biochemistry assessments at Assessment Visits M3 and M6 and adjustments in footnote 7 to correct study procedures regarding patient reported data collection.

The updated Schedule of Assessments is shown overleaf.

Schedule of Assessments: Screening, Randomization and Treatment Phase (Week 0 to Week 10)

Visit Visit Window (±2 days, if applicable)	**Screening** ≤28 days prior to start of IMP	**Rando-mization** Not a patient visit	**Treatment Visit 1** Day 1, Cycle 1 Week 0 Month 0	**Treatment Visit 2** Day 1, Cycle 2 Week 2	**Treatment Visit 3** Day 1, Cycle 3 Week 4	**Treatment Visit 4** Day 1, Cycle 4 Week 6	**Treatment Visit 5** Day 1, Cycle 5 Week 8	**Treatment Visit 6** Day 1, Cycle 6 Week 10
Informed Consent	X							
Inclusion/Exclusion Criteria		X						
CT/MRI Scan and Disease Assessment[1]	X							
CEA	X (within 7 days prior to rando-mization)							
Medical History and Prior Medication	X (within 7 days prior to rando-mization)							
Concomitant Medication			X	X	X	X	X	X
Physical Examination (per standard of care)	X (within 7 days prior to rando-mization)							
Vital Signs (all Visits) and Weight (only for Treatment Visits)	X (within 7 days prior to rando-mization)		X before IMP infusion	X before IMP infusion	X before IMP infusion	X before IMP infusion	X before IMP infusion	X before IMP infusion
ECOG Performance Status	X (within 7 days prior to rando-mization)		X before IMP infusion	X before IMP infusion	X before IMP infusion	X before IMP infusion	X before IMP infusion	X before IMP infusion
Pregnancy Test[2]	X		X		X		X	

Visit Visit Window (±2 days, if applicable)	**Screening** ≤28 days prior to start of IMP	**Rando-mization** Not a patient visit	**Treatment Visit 1** Day 1, Cycle 1 Week 0 Month 0	**Treatment Visit 2** Day 1, Cycle 2 Week 2	**Treatment Visit 3** Day 1, Cycle 3 Week 4	**Treatment Visit 4** Day 1, Cycle 4 Week 6	**Treatment Visit 5** Day 1, Cycle 5 Week 8	**Treatment Visit 6** Day 1, Cycle 6 Week 10
	(within 7 days prior to rando-mization)		before IMP infusion		before IMP infusion		before IMP infusion	
Demographics	X (within 7 days prior to rando-mization)							
Hematology[3]	X (within 7 days prior to rando-mization)		X before IMP infusion	X before IMP infusion	X before IMP infusion	X before IMP infusion	X before IMP infusion	X before IMP infusion
Biochemistry[4]	X (within 7 days prior to rando-mization)		X before IMP infusion		X before IMP infusion		X before IMP infusion	
Blood Mn[5]	X Sample taken within 14 days prior to rando-mization after ICF					X before IMP infusion		
ECG	X (within 7 days prior to rando-mization)							
Randomization[6]		X						

Visit Visit Window (±2 days, if applicable)	**Screening** ≤28 days prior to start of IMP	**Rando-mization** Not a patient visit	**Treatment Visit 1** Day 1, Cycle 1 Week 0 Month 0	**Treatment Visit 2** Day 1, Cycle 2 Week 2	**Treatment Visit 3** Day 1, Cycle 3 Week 4	**Treatment Visit 4** Day 1, Cycle 4 Week 6	**Treatment Visit 5** Day 1, Cycle 5 Week 8	**Treatment Visit 6** Day 1, Cycle 6 Week 10
FACT/GOG-NTX-13[7]			X before IMP infusion	X before IMP infusion	X before IMP infusion	X Before IMP infusion	X before IMP infusion	X before IMP infusion
Cold Sensitivity Patient Questionnaire (Paper Diary)	X (within 7 days prior to rando-mization)			X before IMP infusion and Day 1, 2, 3 after IMP infusion		X before IMP infusion and Day 1, 2, 3 after IMP infusion		
Grooved Pegboard			X before IMP infusion	X before ~~study drug~~ **IMP** infusion	X before ~~study drug~~ **IMP** infusion	X before ~~study drug~~ **IMP** infusion	X before ~~study drug~~ **IMP** infusion	X before ~~study drug~~ **IMP** infusion
Vibration Sensitivity Test (Graduated Tuning Fork)			X before IMP infusion	X before ~~study drug~~ **IMP** infusion	X before ~~study drug~~ **IMP** infusion	X before ~~study drug~~ **IMP** infusion	X before ~~study drug~~ **IMP** infusion	X before ~~study drug~~ **IMP** infusion
QoL/Health Status (EQ-5D-5L)[7]			X before IMP infusion			X before IMP infusion		
Pain NRS[7]			X Before IMP infusion	X Before IMP infusion	X Before IMP infusion	X Before IMP infusion	X Before IMP infusion	X Before IMP infusion
IMP (PledOx or Placebo) Administration[8]			X Only on Day 1, 10 min before chemotherapy infusion	X Only on Day 1, 10 min before chemotherapy infusion	X Only on Day 1, 10 min before chemotherapy infusion	X Only on Day 1, 10 min before chemotherapy infusion	X Only on Day 1, 10 min before chemotherapy infusion	X Only on Day 1, 10 min before chemotherapy infusion
AEs[9]			X	X	X	X	X	X

Visit Visit Window (±2 days, if applicable)	**Screening** ≤28 days prior to start of IMP	**Rando-mization** Not a patient visit	**Treatment Visit 1** Day 1, Cycle 1 Week 0 Month 0	**Treatment Visit 2** Day 1, Cycle 2 Week 2	**Treatment Visit 3** Day 1, Cycle 3 Week 4	**Treatment Visit 4** Day 1, Cycle 4 Week 6	**Treatment Visit 5** Day 1, Cycle 5 Week 8	**Treatment Visit 6** Day 1, Cycle 6 Week 10
MRI of CNS and Blood Mn incl. Neurological Exam[10]			(X)[10]	(X)[10]	(X)[10]	(X)[10]	(X)[10]	(X)[10]
Health Economic Impact[12]			X					
DFS								

[1] CT-scan or MRI, but the same assessment should be performed throughout the study for each patient. For screening, results from a scan performed within 42 days prior to surgery or after surgery are allowed. Tumors will be analyzed and records will be evaluated locally.
[2] WOCBP must give a negative serum pregnancy test at screening and then every 4 weeks while patient is on treatment
[3] Hematology (every Treatment Visit): WBC count differential, ANC, RBC count, platelet count, and hemoglobin. Analyzed by a local hospital laboratory.
[4] Biochemistry (every 2nd Treatment Visit): ALP, total bilirubin, ALT, AST, serum creatinine, and albumin. Analyzed by a local hospital laboratory.
[5] Blood Mn (full blood): Collected at Treatment Visit 4, 8, 12 and EOT. Mn samples will also be taken whenever Parkinson-like symptoms occur. Analyzed by a central laboratory.
[6] Randomization – this is not a patient visit. This is the moment when the Investigator will confirm that a patient complies with all inclusion/exclusion criteria and the patient will be randomized into a treatment arm using the electronic randomization system within eCRF. The Investigator will then request the unblinded pharmacist/designee to prepare the study treatment according to the randomization schedule available only to the unblinded hospital staff.
[7] Questionnaire to be completed electronically **prior to chemotherapy treatment** by a patient on a tablet in the hospital. ~~If a patient does not attend the hospital for the Assessment Visit, the study staff will conduct a recorded telephone call with the patient to collect the relevant data.~~
[8] IMP (PledOx or placebo) will be administered before the background chemotherapy (mFOLFOX6). Study treatment will be stopped in case of mFOLFOX6 treatment discontinuation, disease progression, intolerable toxicity or withdrawal of consent.
[9] All AEs including SAEs will be collected up to 30 days after the last administration of IMP and followed until resolution.
[10] In patients who develop Parkinson-like symptoms, a neurological examination and an MRI of the CNS will be performed and an additional blood Mn sample will be analyzed. An MRI of the CNS should also be performed if elevated Mn is found during the scheduled Mn samples.
[11] The Assessment Visit at Month 6 will be combined with the EOT Visit only if a patient had the last study treatment dose on Treatment Visit 12 (Week 22).
[12] The Health Economic Impact Questionnaire will be completed by the Investigator.

Schedule of Assessments: Treatment Phase (Month 3 to Month ≥6) and End of Treatment Visit

Visit Visit Window (±2 days, if applicable)	**Assessment Visit** Month 3	**Treatment Visit 7** Day 1, Cycle 7 Week 12	**Treatment Visit 8** Day 1, Cycle 8 Week 14	**Treatment Visit 9** Day 1, Cycle 9 Week 16	**Treatment Visit 10** Day 1, Cycle 10 Week 18	**Treatment Visit 11** Day 1, Cycle 11 Week 20	**Treatment Visit 12** Day 1, Cycle 12 Week 22	**Assessment Visit** Month 6[11]	**EOT Visit** Day 14 **(± 3 days)**, Cycle 12 (or Last Cycle)
Informed Consent									
Inclusion/Exclusion Criteria									
CT/MRI Scan and Disease Assessment[1]									
CEA									X
Medical History and Prior Medication									
Concomitant Medication	X	X	X	X	X	X	X	X	X
Physical Examination (per Standard of Care)	X							X	X
Vital Signs (all Visits) and Weight (only for Treatment Visits and EOT visit)	X	X before IMP infusion	X before IMP infusion	X before IMP infusion	X before IMP infusion	X before IMP infusion	X before IMP infusion	X	X
ECOG Performance Status	X	X before IMP infusion	X before IMP infusion	X before IMP infusion	X before IMP infusion	X before IMP infusion	X before IMP infusion	X	X
Pregnancy Test[2]		X before IMP infusion		X before IMP infusion		X before IMP infusion			X
Demographics									
Hematology[3]	X	X before IMP infusion	X before IMP infusion	X before IMP infusion	X before IMP infusion	X before IMP infusion	X before IMP infusion	X	X

Visit Visit Window (±2 days, if applicable)	**Assessment Visit** Month 3	**Treatment Visit 7** Day 1, Cycle 7 Week 12	**Treatment Visit 8** Day 1, Cycle 8 Week 14	**Treatment Visit 9** Day 1, Cycle 9 Week 16	**Treatment Visit 10** Day 1, Cycle 10 Week 18	**Treatment Visit 11** Day 1, Cycle 11 Week 20	**Treatment Visit 12** Day 1, Cycle 12 Week 22	**Assessment Visit** Month 6[11]	**EOT Visit** Day 14 **(± 3 days)**, Cycle 12 (or Last Cycle)
Biochemistry[4]	~~X~~	X before IMP infusion		X before IMP infusion		X before IMP infusion		~~X~~	X
Blood Mn[5]			X before IMP infusion				X before IMP infusion		X
ECG									
Randomization[6]									
FACT/GOG-NTX-13[7]	X	X before IMP infusion	X before IMP infusion	X before IMP infusion	X before IMP infusion	X before IMP infusion	X before IMP infusion	X	X
Cold Sensitivity Patient Questionnaire (Paper Diary)			X before IMP infusion and Day 1, 2, 3 after IMP infusion						
Grooved Pegboard	X	X before IMP infusion	X before IMP infusion	X before IMP infusion	X before IMP infusion	X before IMP infusion	X before IMP infusion	X	X
Vibration Sensitivity Test (Graduated Tuning Fork)	X	X before IMP infusion	X before IMP infusion	X before IMP infusion	X before IMP infusion	X before IMP infusion	X before IMP infusion	X	X
QoL/Health Status (EQ-5D-5L) [7]	X		X before IMP infusion				X before IMP infusion	X	X
Pain NRS[7]	X	X before IMP infusion	X before IMP infusion	X before IMP infusion	X before IMP infusion	X before IMP infusion	X before IMP infusion	X	X

Visit Visit Window (±2 days, if applicable)	**Assessment Visit** Month 3	**Treatment Visit 7** Day 1, Cycle 7 Week 12	**Treatment Visit 8** Day 1, Cycle 8 Week 14	**Treatment Visit 9** Day 1, Cycle 9 Week 16	**Treatment Visit 10** Day 1, Cycle 10 Week 18	**Treatment Visit 11** Day 1, Cycle 11 Week 20	**Treatment Visit 12** Day 1, Cycle 12 Week 22	**Assessment Visit** Month 6[11]	**EOT Visit** Day 14 **(± 3 days)**, Cycle 12 (or Last Cycle)
IMP (PledOx or Placebo) Administration[8]		X 10 min before chemotherapy infusion only on Day 1	X 10 min before chemotherapy infusion only on Day 1	X 10 min before chemotherapy infusion only on Day 1	X 10 min before chemotherapy infusion only on Day 1	X 10 min before chemotherapy infusion only on Day 1	X 10 min before chemotherapy infusion only on Day 1		
AEs[9]		X	X	X	X	X	X		X
MRI of CNS and Blood Mn incl. Neurological Exam[10]	(X)[10]	(X)[10]	(X)[10]	(X)[10]	(X)[10]	(X)[10]	(X)[10]		(X)[10]
Health Economic Impact[12]								X	
DFS	X							X	

[1] CT-scan or MRI, but the same assessment should be performed throughout the study for each patient. Tumors will be analyzed and the records will be evaluated locally.
[2] WOCBP must give a negative serum pregnancy test at screening and then every 4 weeks while patient is on treatment.
[3] Hematology (every Treatment Visit): WBC count differential, ANC, RBC count, platelet count, and hemoglobin. Analyzed by a local hospital laboratory.
[4] Biochemistry (every 2nd Treatment Visit): ALP, total bilirubin, ALT, AST, serum creatinine, and albumin. Analyzed by a local hospital laboratory.
[5] Blood Mn (full blood): Collected at Treatment Visit 4, 8, 12 and EOT. Mn samples will also be taken whenever Parkinson-like symptoms occur. Analyzed by a central laboratory.
[6] Randomization – this is not a patient visit. This is the moment when the Investigator will confirm that a patient complies with all inclusion/exclusion criteria and the patient will be randomized into a treatment arm using the electronic randomization system within eCRF. The Investigator will then request the unblinded pharmacist/designee to prepare the study treatment according to the randomization schedule available only to the unblinded hospital staff.
[7] Questionnaire to be completed electronically **prior to chemotherapy treatment** by a patient on a tablet in the hospital. ~~If a patient does not attend the hospital for the Assessment Visit, the study staff will conduct a recorded telephone call with the patient to collect the relevant data.~~
[8] IMP (PledOx or placebo) will be administered before the background chemotherapy (mFOLFOX6). Study treatment will be stopped in case of mFOLFOX6 treatment discontinuation, disease progression, intolerable toxicity or withdrawal of consent.
[9] All AEs including SAEs will be collected up to 30 days after the last administration of IMP and followed until resolution.
[10] In patients who develop Parkinson-like symptoms, a neurological examination and an MRI of the CNS will be performed and an additional blood Mn sample will be analyzed. An MRI of the CNS should also be performed if elevated Mn is found during the scheduled Mn samples.
[11] The Assessment Visit at Month 6 will be combined with the EOT Visit only if a patient had the last study treatment dose on Treatment Visit 12 (Week 22).
[12] The Health Economic Impact Questionnaire will be completed by the Investigator.

Schedule of Assessments: Follow-up Phase (Month 9 to Month 24 [EOS])

Visit Visit Window (±1 week)	**Assessment Visit** Week 36 Month 9	**Assessment Visit** Week 48 Month 12	**Assessment Visit** Week 72 Month 18	**EOS Visit** Week 96 Month 24
Informed Consent				
Inclusion/Exclusion Criteria				
CT/MRI Scan and Disease Assessment[1]		X		X
CEA		X		X
Medical History and Prior Medication				
Concomitant Medication	X	X	X	X
Physical Examination (per Standard of Care)	X	X	X	X
Vital Signs (all Visits) and Weight (only for Treatment Visit)	X	X	X	X
ECOG Performance Status	X	X	X	X
Pregnancy Test[2]				
Demographics				
Hematology[3]				
Biochemistry[4]				
Blood Mn[5]				
ECG				
Randomization[6]				
FACT/GOG-NTX-13[7]	X	X	X	X

Visit Visit Window (±1 week)	**Assessment Visit** Week 36 Month 9	**Assessment Visit** Week 48 Month 12	**Assessment Visit** Week 72 Month 18	**EOS Visit** Week 96 Month 24
Cold Sensitivity Patient Questionnaire (Paper Diary)				
Grooved Pegboard	X	X	X	X
Vibration Sensitivity Test (Graduated Tuning Fork)	X	X	X	X
QoL/Health Status[7] (EQ-5D-5L)	X	X	X	X
Pain NRS[7]	X	X	X	X
IMP (PledOx or Placebo) Administration[8]				
AEs[9]				
MRI of CNS and Blood Mn incl. Neurological Exam[10]				
Health Economic Impact[12]		X	X	X
DFS	X	X	X	X

[1] CT-scan or MRI, but the same assessment should be performed throughout the study for each patient. Tumors will be analyzed and the records will be evaluated locally.
[2] WOCBP must give a negative serum pregnancy test at screening and then every 4 weeks while patient is on treatment.
[3] Hematology (every Treatment Visit): WBC count differential, ANC, RBC count, platelet count, and hemoglobin. Analyzed by a local hospital laboratory.
[4] Biochemistry (every 2nd Treatment Visit): ALP, total bilirubin, ALT, AST, serum creatinine, and albumin. Analyzed by a local hospital laboratory.
[5] Blood Mn (full blood): Collected at Treatment Visit 4, 8, 12 and EOT. Mn samples will also be taken whenever Parkinson-like symptoms occur. Analyzed by a central laboratory.
[6] Randomization – this is not a patient visit. This is the moment when the Investigator will confirm that a patient complies with all inclusion/exclusion criteria and the patient will be randomized into a treatment arm using the electronic randomization system within eCRF. The Investigator will then request the unblinded pharmacist/designee to prepare the study treatment according to the randomization schedule available only to the unblinded hospital staff.
[7] Questionnaire to be completed electronically **prior to chemotherapy treatment** by a patient on a tablet in the hospital. ~~If a patient does not attend the hospital for the Assessment Visit, the study staff will conduct a recorded telephone call with the patient to collect the relevant data.~~
[8] IMP (PledOx or placebo) will be administered before the background chemotherapy (mFOLFOX6). Study treatment will be stopped in case of mFOLFOX6 treatment discontinuation, disease progression, intolerable toxicity or withdrawal of consent.
[9] All AEs including SAEs will be collected up to 30 days after the last administration of IMP and followed until resolution.
[10] In patients who develop Parkinson-like symptoms, a neurological examination and an MRI of the CNS will be performed and an additional blood Mn sample will be analyzed. An MRI of the CNS should also be performed if elevated Mn is found during the scheduled Mn samples.

[11] The Assessment Visit at Month 6 will be combined with the EOT Visit only if a patient had the last study treatment dose on Treatment Visit 12 (Week 22).
[12] The Health Economic Impact Questionnaire will be completed by the Investigator.

3. APPENDICES

3.1 APPENDIX 1: STUDY ACKNOWLEDGEMENT

Investigator's Statement:

I have read and understand the foregoing Protocol entitled "A Phase 3, double-blind, multicenter, placebo-controlled study of PledOx used on top of Modified FOLFOX6 (5-FU/FA and Oxaliplatin) to prevent chemotherapy induced peripheral neuropathy (CIPN) in the adjuvant treatment of patients with Stage III or high-risk Stage II colorectal cancer", protocol number PP06489 (POLAR-A), and agree to conduct the study in compliance with Good Clinical Practice (CPMP/ICH/135/95), ICH GCP, designated standard operating procedures, national laws and regulations of the countries conducting the study and within the principles of the Declaration of Helsinki as outlined herein.

Investigator's Name (please print)	Investigator's Title
Date (dd mmm yyyy)	Investigator's Signature